

Georgia-All Variables-Complete Analysis-UPDATED (1)

May 18, 2026

```
[47]: library(dplyr)
library(tidyr)
library(forcats)
library(car)
library(caret)
library(vcd)
library(pscl)
library(glmnet)
library(gglasso)
library(ggplot2)
library(ResourceSelection)
library(detectseparation)
library(Matrix)
library(pROC)
library(tibble)
georgia <- read.csv("Georgia Dataset.csv", header=TRUE)
#georgia <- read.csv("Georgia Dataset.csv", header=TRUE)
head(georgia)
str(georgia)
```

	ID	Gender	Race	Age_at_Release	Residence_PUMA	Gang_Affiliated
	<int>	<chr>	<chr>	<chr>	<int>	<chr>
1	1	M	BLACK	43-47	16	false
2	2	M	BLACK	33-37	16	false
3	3	M	BLACK	48 or older	24	false
4	4	M	WHITE	38-42	16	false
5	5	M	WHITE	33-37	16	false
6	6	M	WHITE	38-42	17	false

'data.frame': 25835 obs. of 54 variables:

```
$ ID : int 1 2 3 4 5 6 7 8 9 10
...
$ Gender : chr "M" "M" "M" "M" ...
$ Race : chr "BLACK" "BLACK"
"BLACK" "WHITE" ...
$ Age_at_Release : chr "43-47" "33-37" "48
or older" "38-42" ...
$ Residence_PUMA : int 16 16 24 16 16 17 18
```

```

16 5 16 ...
$ Gang_Affiliated : chr "false" "false"
"false" "false" ...
$ Supervision_Risk_Score_First : int 3 6 7 7 4 5 2 5 7 5
...
$ Supervision_Level_First : chr "Standard"
"Specialized" "High" "High" ...
$ Education_Level : chr "At least some
college" "Less than HS diploma" "At least some college" "Less than HS diploma"
...
$ Dependents : chr "3 or more" "1" "3 or
more" "1" ...
$ Prison_Offense : chr "Drug" "Violent/Non-
Sex" "Drug" "Property" ...
$ Prison_Years : chr "More than 3 years"
"More than 3 years" "1-2 years" "1-2 years" ...
$ Prior_Arrest_Episodes_Felony : chr "6" "7" "6" "8" ...
$ Prior_Arrest_Episodes_Misd : chr "6 or more" "6 or
more" "6 or more" "6 or more" ...
$ Prior_Arrest_Episodes_Violent : chr "1" "3 or more" "3 or
more" "0" ...
$ Prior_Arrest_Episodes_Property : chr "3" "0" "2" "3" ...
$ Prior_Arrest_Episodes_Drug : chr "3" "3" "2" "3" ...
$ Prior_Arrest_Episodes_PPViolationCharges : chr "4" "5 or more" "5 or
more" "3" ...
$ Prior_Arrest_Episodes_DVCharges : chr "false" "true" "true"
"false" ...
$ Prior_Arrest_Episodes_GunCharges : chr "false" "false"
"false" "false" ...
$ Prior_Conviction_Episodes_Felony : chr "3 or more" "3 or
more" "3 or more" "3 or more" ...
$ Prior_Conviction_Episodes_Misd : chr "3" "4 or more" "2"
"4 or more" ...
$ Prior_Conviction_Episodes_Viol : chr "false" "true" "true"
"false" ...
$ Prior_Conviction_Episodes_Prop : chr "2" "0" "1" "3 or
more" ...
$ Prior_Conviction_Episodes_Drug : chr "2 or more" "2 or
more" "2 or more" "2 or more" ...
$ Prior_Conviction_Episodes_PPViolationCharges : chr "false" "true"
"false" "false" ...
$ Prior_Conviction_Episodes_DomesticViolenceCharges : chr "false" "true" "true"
"false" ...
$ Prior_Conviction_Episodes_GunCharges : chr "false" "true"
"false" "false" ...
$ Prior_Revocations_Parole : chr "false" "false"
"false" "false" ...
$ Prior_Revocations_Probation : chr "false" "false"

```

```

"false" "true" ...
$ Condition_MH_SA : chr "true" "false" "true"
"true" ...
$ Condition_Cog_Ed : chr "true" "false" "true"
"true" ...
$ Condition_Other : chr "false" "false"
"false" "false" ...
$ Violations_ElectronicMonitoring : chr "false" "false"
"false" "false" ...
$ Violations_Instruction : chr "false" "true" "true"
"false" ...
$ Violations_FailToReport : chr "false" "false"
"false" "false" ...
$ Violations_MoveWithoutPermission : chr "false" "false"
"true" "false" ...
$ Delinquency_Reports : chr "0" "4 or more" "4 or
more" "0" ...
$ Program_Attendances : chr "6" "0" "6" "6" ...
$ Program_UnexcusedAbsences : chr "0" "0" "0" "0" ...
$ Residence_Changes : chr "2" "2" "0" "3 or
more" ...
$ Avg_Days_per_DrugTest : num 612 35.7 93.7 25.4
23.1 ...
$ DrugTests_THC_Positive : num 0 0 0.333 0 0 ...
$ DrugTests_Cocaine_Positive : num 0 0 0 0 0 0 0 0 NA 0
...
$ DrugTests_Meth_Positive : num 0 0 0.1667 0 0.0588
...
$ DrugTests_Other_Positive : num 0 0 0 0 0 0 0 0 NA 0
...
$ Percent_Days_Employed : num 0.489 0.425 0 1 0.204
...
$ Jobs_Per_Year : num 0.448 2 0 0.719 0.929
...
$ Employment_Exempt : chr "false" "false"
"false" "false" ...
$ Recidivism_Within_3years : chr "false" "true" "true"
"false" ...
$ Recidivism_Arrest_Year1 : chr "false" "false"
"false" "false" ...
$ Recidivism_Arrest_Year2 : chr "false" "false"
"true" "false" ...
$ Recidivism_Arrest_Year3 : chr "false" "true"
"false" "false" ...
$ Training_Sample : int 1 1 1 1 1 0 1 0 1 1
...

```

```
[2]: #check for duplicate IDs - an empty table means there are no duplicates. we
      ↪want that!
n_occur <- data.frame(table(georgia$ID))
n_occur[n_occur$Freq > 1,]
```

A data.frame: 0 × 2 Var1 Freq
 <fct> <int>

```
[3]: #some column type cleaning

#convert true/false columns to 1 and 0
for (colname in names(georgia)) {
  if (is.character(georgia[[colname]])) {
    lower_vals <- tolower(georgia[[colname]])
    lower_vals[lower_vals == ""] <- NA # Convert blank strings to NA
    unique_vals <- unique(na.omit(lower_vals))

    if (setequal(unique_vals, c("true", "false"))) {
      georgia[[colname]] <- factor(lower_vals, levels = c("false", "true"),
      ↪labels = c(0, 1))
    }
  }
}

#a bunch of as.factor!!!!
georgia$Gender <- as.factor(georgia$Gender)
georgia$Race <- as.factor(georgia$Race)
georgia$Age_at_Release <- as.factor(georgia$Age_at_Release)
georgia$Supervision_Level_First <- as.factor(georgia$Supervision_Level_First)
georgia$Education_Level <- as.factor(georgia$Education_Level)
georgia$Dependents <- as.factor(georgia$Dependents)
georgia$Prison_Offense <- as.factor(georgia$Prison_Offense)
georgia$Prison_Years <- as.factor(georgia$Prison_Years)
georgia$Prior_Arrest_Episodes_Felony <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_Felony)
georgia$Prior_Arrest_Episodes_Misd <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_Misd)
georgia$Prior_Arrest_Episodes_Violent <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_Violent)
georgia$Prior_Arrest_Episodes_Property <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_Property)
georgia$Prior_Arrest_Episodes_Drug <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_Drug)
georgia$Prior_Arrest_Episodes_PPViolationCharges <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_PPViolationCharges)
georgia$Prior_Arrest_Episodes_DVCharges <- as.
  ↪factor(georgia$Prior_Arrest_Episodes_DVCharges)
```

```

georgia$Prior_Arrest_Episodes_GunCharges <- as.
  ↳factor(georgia$Prior_Arrest_Episodes_GunCharges)
georgia$Prior_Conviction_Episodes_Felony <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_Felony)
georgia$Prior_Conviction_Episodes_Misd <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_Misd)
georgia$Prior_Conviction_Episodes_Viol <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_Viol)
georgia$Prior_Conviction_Episodes_Prop <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_Prop)
georgia$Prior_Conviction_Episodes_Drug <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_Drug)
georgia$Prior_Conviction_Episodes_PPViolationCharges <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_PPViolationCharges)
georgia$Prior_Conviction_Episodes_DomesticViolenceCharges <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_DomesticViolenceCharges)
georgia$Prior_Conviction_Episodes_GunCharges <- as.
  ↳factor(georgia$Prior_Conviction_Episodes_GunCharges)
georgia$Delinquency_Reports <- as.factor(georgia$Delinquency_Reports)
georgia$Program_Attendances <- as.factor(georgia$Program_Attendances)
georgia$Program_UnexcusedAbsences <- as.
  ↳factor(georgia$Program_UnexcusedAbsences)
georgia$Residence_Changes <- as.factor(georgia$Residence_Changes)

#define a within 2 year recidivism for analysis; matches wisconsin
georgia$Recidivism_Within_2years <- as.
  ↳factor(ifelse(georgia$Recidivism_Arrest_Year1 == "1"
  |
  ↳georgia$Recidivism_Arrest_Year2 == "1", 1, 0))

#relabel the levels of the race column to conform with wisconsin data
georgia$Race <- fct_recode(georgia$Race, "Caucasian" = "WHITE", "African_
  ↳American" = "BLACK")

#everything nice and clean
head(georgia)
str(georgia)

```

A data.frame: 6 × 55

	ID	Gender	Race	Age_at_Release	Residence_PUMA	Gang_A
	<int>	<fct>	<fct>	<fct>	<int>	<fct>
1	1	M	African American	43-47	16	0
2	2	M	African American	33-37	16	0
3	3	M	African American	48 or older	24	0
4	4	M	Caucasian	38-42	16	0
5	5	M	Caucasian	33-37	16	0
6	6	M	Caucasian	38-42	17	0

```

'data.frame':  25835 obs. of  55 variables:
 $ ID                                     : int  1 2 3 4 5 6 7 8 9 10
...
 $ Gender                               : Factor w/ 2 levels
"F","M": 2 2 2 2 2 2 2 2 1 2 ...
 $ Race                                 : Factor w/ 2 levels
"African American",...: 1 1 1 2 2 2 1 1 1 1 ...
 $ Age_at_Release                       : Factor w/ 7 levels
"18-22","23-27",...: 6 4 7 5 4 5 7 5 6 6 ...
 $ Residence_PUMA                       : int  16 16 24 16 16 17 18
16 5 16 ...
 $ Gang_Affiliated                     : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 1 1 NA 1 ...
 $ Supervision_Risk_Score_First         : int  3 6 7 7 4 5 2 5 7 5
...
 $ Supervision_Level_First              : Factor w/ 4 levels
","High","Specialized",...: 4 3 2 2 3 4 4 2 2 4 ...
 $ Education_Level                     : Factor w/ 3 levels "At
least some college",...: 1 3 1 3 3 2 3 2 2 2 ...
 $ Dependents                          : Factor w/ 4 levels
"0","1","2","3 or more": 4 2 4 2 4 1 3 4 1 4 ...
 $ Prison_Offense                      : Factor w/ 6 levels
","Drug","Other",...: 2 5 2 4 5 4 1 2 2 4 ...
 $ Prison_Years                        : Factor w/ 4 levels "1-2
years","Greater than 2 to 3 years",...: 4 4 1 1 1 4 3 2 3 4 ...
 $ Prior_Arrest_Episodes_Felony        : Factor w/ 11 levels
"0","1","10 or more",...: 8 9 8 10 6 6 3 8 3 3 ...
 $ Prior_Arrest_Episodes_Misd          : Factor w/ 7 levels
"0","1","2","3",...: 7 7 7 7 5 1 7 7 5 7 ...
 $ Prior_Arrest_Episodes_Violent       : Factor w/ 4 levels
"0","1","2","3 or more": 2 4 4 1 4 2 2 4 3 4 ...
 $ Prior_Arrest_Episodes_Property      : Factor w/ 6 levels
"0","1","2","3",...: 4 1 3 4 3 4 6 2 6 6 ...
 $ Prior_Arrest_Episodes_Drug          : Factor w/ 6 levels
"0","1","2","3",...: 4 4 3 4 2 1 2 3 4 3 ...
 $ Prior_Arrest_Episodes_PPViolationCharges : Factor w/ 6 levels
"0","1","2","3",...: 5 6 6 4 4 1 6 6 4 6 ...
 $ Prior_Arrest_Episodes_DVCharges     : Factor w/ 2 levels
"0","1": 1 2 2 1 2 1 1 1 1 2 ...
 $ Prior_Arrest_Episodes_GunCharges    : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 2 1 1 2 ...
 $ Prior_Conviction_Episodes_Felony    : Factor w/ 4 levels
"0","1","2","3 or more": 4 4 4 4 2 2 4 2 3 4 ...
 $ Prior_Conviction_Episodes_Misd      : Factor w/ 5 levels
"0","1","2","3",...: 4 5 3 5 1 1 2 5 5 5 ...
 $ Prior_Conviction_Episodes_Viol     : Factor w/ 2 levels
"0","1": 1 2 2 1 2 1 1 2 1 2 ...
 $ Prior_Conviction_Episodes_Prop      : Factor w/ 4 levels

```

```

"0","1","2","3 or more": 3 1 2 4 1 3 4 1 4 4 ...
$ Prior_Conviction_Episodes_Drug : Factor w/ 3 levels
"0","1","2 or more": 3 3 3 3 2 1 1 3 3 3 ...
$ Prior_Conviction_Episodes_PPViolationCharges : Factor w/ 2 levels
"0","1": 1 2 1 1 1 1 2 2 2 2 ...
$ Prior_Conviction_Episodes_DomesticViolenceCharges: Factor w/ 2 levels
"0","1": 1 2 2 1 1 1 1 1 1 2 ...
$ Prior_Conviction_Episodes_GunCharges : Factor w/ 2 levels
"0","1": 1 2 1 1 1 1 2 1 1 2 ...
$ Prior_Revocations_Parole : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ Prior_Revocations_Probation : Factor w/ 2 levels
"0","1": 1 1 1 2 1 1 1 1 2 1 ...
$ Condition_MH_SA : Factor w/ 2 levels
"0","1": 2 1 2 2 2 1 1 1 2 1 ...
$ Condition_Cog_Ed : Factor w/ 2 levels
"0","1": 2 1 2 2 2 1 1 2 1 2 ...
$ Condition_Other : Factor w/ 2 levels
"0","1": 1 1 1 1 2 2 1 1 1 2 ...
$ Violations_ElectronicMonitoring : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ Violations_Instruction : Factor w/ 2 levels
"0","1": 1 2 2 1 1 1 1 1 1 1 ...
$ Violations_FailToReport : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ Violations_MoveWithoutPermission : Factor w/ 2 levels
"0","1": 1 1 2 1 1 2 1 1 1 1 ...
$ Delinquency_Reports : Factor w/ 5 levels
"0","1","2","3",...: 1 5 5 1 1 1 1 1 1 1 ...
$ Program_Attendances : Factor w/ 11 levels
"0","1","10 or more",...: 8 1 8 8 9 1 1 1 1 1 ...
$ Program_UnexcusedAbsences : Factor w/ 4 levels
"0","1","2","3 or more": 1 1 1 1 1 1 1 1 1 1 ...
$ Residence_Changes : Factor w/ 4 levels
"0","1","2","3 or more": 3 3 1 4 1 4 2 1 1 1 ...
$ Avg_Days_per_DrugTest : num 612 35.7 93.7 25.4
23.1 ...
$ DrugTests_THC_Positive : num 0 0 0.333 0 0 ...
$ DrugTests_Cocaine_Positive : num 0 0 0 0 0 0 0 0 NA 0
...
$ DrugTests_Meth_Positive : num 0 0 0.1667 0 0.0588
...
$ DrugTests_Other_Positive : num 0 0 0 0 0 0 0 0 NA 0
...
$ Percent_Days_Employed : num 0.489 0.425 0 1 0.204
...
$ Jobs_Per_Year : num 0.448 2 0 0.719 0.929
...

```

```

$ Employment_Exempt           : Factor w/ 2 levels
"0","1": 1 1 1 1 1 1 1 2 1 ...
$ Recidivism_Within_3years     : Factor w/ 2 levels
"0","1": 1 2 2 1 2 1 2 1 2 2 ...
$ Recidivism_Arrest_Year1      : Factor w/ 2 levels
"0","1": 1 1 1 1 2 1 1 1 2 2 ...
$ Recidivism_Arrest_Year2      : Factor w/ 2 levels
"0","1": 1 1 2 1 1 1 1 1 1 1 ...
$ Recidivism_Arrest_Year3      : Factor w/ 2 levels
"0","1": 1 2 1 1 1 1 2 1 1 1 ...
$ Training_Sample              : int   1 1 1 1 1 0 1 0 1 1
...
$ Recidivism_Within_2years     : Factor w/ 2 levels
"0","1": 1 1 2 1 2 1 1 1 2 2 ...

```

```

[4]: # Replace blank strings "" with NA
georgia$Supervision_Level_First[georgia$Supervision_Level_First == ""] <- NA
georgia$Prison_Offense[georgia$Prison_Offense == ""] <- NA

#replace NAs with "Unknown"
georgia$Gang_Affiliated <- fct_na_value_to_level(georgia$Gang_Affiliated, level_
  ↳ "Unknown")
georgia$Supervision_Level_First <-
  ↳ fct_na_value_to_level(georgia$Supervision_Level_First, level = "Unknown")
georgia$Prison_Offense <- fct_na_value_to_level(georgia$Prison_Offense, level =
  ↳ "Unknown")

#drop unused levels
georgia$Supervision_Level_First <- droplevels(georgia$Supervision_Level_First)
georgia$Prison_Offense <- droplevels(georgia$Prison_Offense)

#impute NAs in int columns with median
georgia <- georgia %>% mutate(across(where(is.numeric), ~ ifelse(is.na(.),
  ↳ median(., na.rm = TRUE), .)))

#remove training sample column; this was added by researchers and not necessary
  ↳ for our analysis
georgia$Training_Sample <- NULL

summary(georgia)

```

	ID	Gender	Race	Age_at_Release
Min.	: 1	F: 3167	African American:14847	18-22 :2066
1st Qu.:	6626	M:22668	Caucasian :10988	23-27 :5176
Median	:13270			28-32 :4982
Mean	:13314			33-37 :4271
3rd Qu.:	20022			38-42 :2993
Max.	:26761			43-47 :2620

48 or older:3727

Residence_PUMA	Gang_Affiliated	Supervision_Risk_Score_First
Min. : 1.00	0 :18730	Min. : 1.000
1st Qu.: 6.00	1 : 3938	1st Qu.: 4.000
Median :12.00	Unknown: 3167	Median : 6.000
Mean :12.36		Mean : 6.081
3rd Qu.:18.00		3rd Qu.: 8.000
Max. :25.00		Max. :10.000

Supervision_Level_First	Education_Level	Dependents
High :7009	At least some college: 4605	0 :8037
Specialized:7123	High School Diploma :11390	1 :5369
Standard :9983	Less than HS diploma : 9840	2 :4647
Unknown :1720		3 or more:7782

Prison_Offense	Prison_Years
Drug :5190	1-2 years :8084
Other :2779	Greater than 2 to 3 years:4207
Property :8284	Less than 1 year :8048
Violent/Non-Sex:5475	More than 3 years :5496
Violent/Sex : 830	
Unknown :3277	

Prior_Arrest_Episodes_Felony	Prior_Arrest_Episodes_Misd
10 or more:6140	0 :4099
3 :2768	1 :3479
4 :2697	2 :3121
2 :2622	3 :2654
5 :2483	4 :2340
1 :2250	5 :1936
(Other) :6875	6 or more:8206

Prior_Arrest_Episodes_Violent	Prior_Arrest_Episodes_Property
0 :11049	0 :6517
1 : 7287	1 :5108
2 : 3613	2 :3886
3 or more: 3886	3 :2653
	4 :1880
	5 or more:5791

Prior_Arrest_Episodes_Drug	Prior_Arrest_Episodes_PPViolationCharges
0 :8002	0 :6699
1 :5562	1 :4346
2 :4162	2 :3449
3 :3042	3 :2775
4 :1993	4 :2249
5 or more:3074	5 or more:6317

Prior_Arrest_Episodes_DVCharges	Prior_Arrest_Episodes_GunCharges
0:21567	0:19010
1: 4268	1: 6825

Prior_Conviction_Episodes_Felony	Prior_Conviction_Episodes_Misd
0 :7539	0 :7819
1 :6963	1 :5387
2 :4374	2 :3888
3 or more:6959	3 :2759
	4 or more:5982

Prior_Conviction_Episodes_Viol	Prior_Conviction_Episodes_Prop
0:17447	0 :11198
1: 8388	1 : 5896
	2 : 3306
	3 or more: 5435

Prior_Conviction_Episodes_Drug	Prior_Conviction_Episodes_PPViolationCharges
0 :12586	0:17367
1 : 6489	1: 8468
2 or more: 6760	

Prior_Conviction_Episodes_DomesticViolenceCharges
0:23726
1: 2109

Prior_Conviction_Episodes_GunCharges	Prior_Revocations_Parole
0:22380	0:23362
1: 3455	1: 2473

Prior_Revocations_Probation	Condition_MH_SA	Condition_Cog_Ed	Condition_Other
0:22068	0: 8827	0:14394	0:17459
1: 3767	1:17008	1:11441	1: 8376

Violations_ElectronicMonitoring	Violations_Instruction	Violations_FailToReport
0:23927	0:21364	0:23881
1: 1908	1: 4471	1: 1954

Violations_MoveWithoutPermission	Delinquency_Reports	Program_Attendances
0:22837	0 :17566	0 :13686
1: 2998	1 : 1229	10 or more: 3231
	2 : 1361	6 : 2809
	3 : 1266	5 : 1142
	4 or more: 4413	1 : 1003
		7 : 830
		(Other) : 3134

Program_UnexcusedAbsences	Residence_Changes	Avg_Days_per_DrugTest
0 :20802	0 :13264	Min. : 0.50
1 : 1682	1 : 6397	1st Qu.: 35.94
2 : 1146	2 : 3322	Median : 55.42
3 or more: 2205	3 or more: 2852	Mean : 84.80
		3rd Qu.: 87.33
		Max. :1088.50

DrugTests_THC_Positive	DrugTests_Cocaine_Positive	DrugTests_Meth_Positive
Min. :0.00000	Min. :0.00000	Min. :0.00000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000
Median :0.00000	Median :0.00000	Median :0.00000
Mean :0.05067	Mean :0.01099	Mean :0.01031
3rd Qu.:0.01732	3rd Qu.:0.00000	3rd Qu.:0.00000
Max. :1.00000	Max. :1.00000	Max. :1.00000

DrugTests_Other_Positive	Percent_Days_Employed	Jobs_Per_Year
Min. :0.000000	Min. :0.0000	Min. :0.0000
1st Qu.:0.000000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.000000	Median :0.4757	Median :0.6352
Mean :0.006039	Mean :0.4822	Mean :0.7651
3rd Qu.:0.000000	3rd Qu.:0.9646	3rd Qu.:1.0000
Max. :1.000000	Max. :1.0000	Max. :8.0000

Employment_Exempt	Recidivism_Within_3years	Recidivism_Arrest_Year1
0: 22399	0: 10931	0: 18111
1: 3436	1: 14904	1: 7724

Recidivism_Arrest_Year2	Recidivism_Arrest_Year3	Recidivism_Within_2years
0: 21268	0: 23222	0: 13544
1: 4567	1: 2613	1: 12291

1 MODEL PREP

```
[5]: set.seed(100) # For reproducibility
train_idx <- sample(1:nrow(georgia), size = 0.8 * nrow(georgia), replace = FALSE)
train_data <- georgia[train_idx, ]
test_data <- georgia[-train_idx, ]
```

```
[6]: table(train_data$Recidivism_Within_2years) / nrow(train_data)
table(test_data$Recidivism_Within_2years) / nrow(test_data)
```

0	1
0.5237081	0.4762919

0	1
0.5264177	0.4735823

```
[7]: #formula focus on Recidivism_Within_2years as response, and removing ID, other
      ↪ recidivism variables
formula <- Recidivism_Within_2years ~ . - ID - Recidivism_Within_3years -
      ↪ Recidivism_Arrest_Year1 - Recidivism_Arrest_Year2 - Recidivism_Arrest_Year3
```

```
[8]: X_all <- model.matrix(formula, data = train_data)[, -1] #remove intercept
y <- as.numeric(as.character(train_data$Recidivism_Within_2years)) #factor to
      ↪ numeric 0/1
```

```
#running a VIF function initially gave an error due to perfect
  ↳ multicollinearity! investigation yielded an interesting result when looking
  ↳ at gender and gang affiliation
alias_model <- alias(lm(y ~ X_all))
#print(alias_model$Complete)
table(georgia$Gender, georgia$Gang_Affiliated)
```

	0	1	Unknown
F	0	0	3167
M	18730	3938	0

```
[9]: #create two models and see which has better AIC; one without gender and one
  ↳ without gang_affiliation
formula_no_gender <- as.formula("Recidivism_Within_2years ~ . - ID -
  ↳ Recidivism_Within_3years - Recidivism_Arrest_Year1 - Recidivism_Arrest_Year2
  ↳ Recidivism_Arrest_Year3 - Gender")
formula_no_gang <- as.formula("Recidivism_Within_2years ~ . - ID -
  ↳ Recidivism_Within_3years - Recidivism_Arrest_Year1 - Recidivism_Arrest_Year2
  ↳ Recidivism_Arrest_Year3 - Gang_Affiliated")

model_no_gender <- glm(formula_no_gender, data = train_data, family = binomial)
model_no_gang <- glm(formula_no_gang, data = train_data, family = binomial)

#compare AIC
AIC(model_no_gender, model_no_gang)

#compare pseudo R² or use cross-validation for more robust comparison
library(pscl)
pR2(model_no_gender)
pR2(model_no_gang)
```

A data.frame: 2 × 2	df	AIC
	<dbl>	<dbl>
model_no_gender	117	23227.62
model_no_gang	116	23436.62

fitting null model for pseudo-r2

llh	-11496.8083574636	llhNull	-14302.7232288885	G2	5611.82974284976	McFadden
	0.196180463434929	r2ML	0.237781966680468	r2CU	0.317280760442644	

fitting null model for pseudo-r2

llh	-11602.3080994425	llhNull	-14302.7232288885	G2	5400.83025889205	McFadden
	0.18880426379165	r2ML	0.229960631449434	r2CU	0.306844480415093	

```
[10]: #model_no_gender slightly better; we'll remove Gender from future analysis
vif_values <- vif(model_no_gender)
```

```
#confirm no multicollinearity issues after removing Gender
print(vif_values)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
Race	1.341399	1	1.158188
Age_at_Release	3.148601	6	1.100297
Residence_PUMA	1.048486	1	1.023956
Gang_Affiliated	1.324906	2	1.072868
Supervision_Risk_Score_First	2.210440	1	1.486755
Supervision_Level_First	2.080885	3	1.129903
Education_Level	1.147708	2	1.035042
Dependents	1.150411	3	1.023628
Prison_Offense	3.957471	5	1.147471
Prison_Years	1.938265	3	1.116612
Prior_Arrest_Episodes_Felony	8.779805	10	1.114742
Prior_Arrest_Episodes_Misd	5.072679	6	1.144906
Prior_Arrest_Episodes_Violent	2.414925	3	1.158290
Prior_Arrest_Episodes_Property	7.134972	5	1.217136
Prior_Arrest_Episodes_Drug	4.327891	5	1.157784
Prior_Arrest_Episodes_PPViolationCharges	4.027414	5	1.149483
Prior_Arrest_Episodes_DVCharges	1.564249	1	1.250700
Prior_Arrest_Episodes_GunCharges	1.552303	1	1.245914
Prior_Conviction_Episodes_Felony	2.796816	3	1.186982
Prior_Conviction_Episodes_Misd	4.196573	4	1.196360
Prior_Conviction_Episodes_Viol	1.908400	1	1.381449
Prior_Conviction_Episodes_Prop	5.251356	3	1.318394
Prior_Conviction_Episodes_Drug	3.293609	2	1.347156
Prior_Conviction_Episodes_PPViolationCharges	1.718642	1	1.310970
Prior_Conviction_Episodes_DomesticViolenceCharges	1.501814	1	1.225485
Prior_Conviction_Episodes_GunCharges	1.557343	1	1.247935
Prior_Revocations_Parole	1.137547	1	1.066559
Prior_Revocations_Probation	1.099561	1	1.048600
Condition_MH_SA	1.274459	1	1.128919
Condition_Cog_Ed	1.267750	1	1.125944
Condition_Other	1.416961	1	1.190362
Violations_ElectronicMonitoring	1.277287	1	1.130171
Violations_Instruction	1.333597	1	1.154815
Violations_FailToReport	1.148764	1	1.071804
Violations_MoveWithoutPermission	1.206799	1	1.098544
Delinquency_Reports	1.557175	4	1.056920
Program_Attendances	1.773756	10	1.029069
Program_UnexcusedAbsences	1.624412	3	1.084217
Residence_Changes	1.117555	3	1.018696
Avg_Days_per_DrugTest	1.101247	1	1.049403
DrugTests_THC_Positive	1.237174	1	1.112283
DrugTests_Cocaine_Positive	1.089779	1	1.043925
DrugTests_Meth_Positive	1.126878	1	1.061545
DrugTests_Other_Positive	1.045141	1	1.022321

Percent_Days_Employed	2.000329	1	1.414330
Jobs_Per_Year	2.026690	1	1.423619
Employment_Exempt	1.068256	1	1.033565

```
[11]: #make groups in prep for group lasso
X_full <- model.matrix(Recidivism_Within_2years ~ . - ID -
  ↳ Recidivism_Within_3years - Recidivism_Arrest_Year1 - Recidivism_Arrest_Year2 -
  ↳ Recidivism_Arrest_Year3 - Gender, data = train_data)
groups_full <- attr(X_full, "assign")

X <- X_full[,-1]
groups <- groups_full[-1]

y <- as.numeric(as.character(train_data$Recidivism_Within_2years)) # Should be
  ↳ 0/1

table(groups) # Shows the number of columns per group
terms_obj <- terms(Recidivism_Within_2years ~ . - ID - Recidivism_Within_3years -
  ↳ Recidivism_Arrest_Year1 - Recidivism_Arrest_Year2 -
  ↳ Recidivism_Arrest_Year3 - Gender, data = train_data)
var_names <- attr(terms_obj, "term.labels")
#print(var_names)
```

```
groups
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
 1  6  1  2  1  3  2  3  5  3 10  6  3  5  5  5  1  1  3  4  1  3  2  1  1  1
27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
 1  1  1  1  1  1  1  1  1  1  4 10  3  3  1  1  1  1  1  1  1  1  1
```

```
[12]: # Exclude variables we don't want (adjust as needed)
exclude_vars <- c("ID", "Recidivism_Within_3years", "Recidivism_Arrest_Year1",
  "Recidivism_Arrest_Year2", "Recidivism_Arrest_Year3",
  ↳ "Gender",
  "Recidivism_Within_2years") # response

#get all predictors
predictor_vars <- setdiff(names(train_data), exclude_vars)

#identify factor and numeric variables
factor_vars <- predictor_vars[sapply(train_data[, predictor_vars], is.factor)]
numeric_vars <- predictor_vars[sapply(train_data[, predictor_vars], is.numeric)]
```

```
[13]: #build the formula
formula <- as.formula(
  paste("~", paste(predictor_vars, collapse = " + "))
)

#create model matrix (no intercept)
```

```
X_full <- model.matrix(formula, data = train_data)[, -1]

#get column names for mapping
col_names <- colnames(X_full)
```

```
[14]: group <- integer(length(col_names))
group_counter <- 1

#assign group numbers for factor variables (all dummies from a factor get the
↳same group)
for (f in factor_vars) {
  matches <- grep(paste0("^", f), col_names)
  group[matches] <- group_counter
  group_counter <- group_counter + 1
}

#assign group numbers for numeric variables (each gets its own group)
for (n in numeric_vars) {
  matches <- which(col_names == n)
  group[matches] <- group_counter
  group_counter <- group_counter + 1
}
```

```
[15]: #response in -1/1 format
Response <- ifelse(train_data$Recidivism_Within_2years == 1, 1, -1)

# Standardize predictors
train_means <- colMeans(X_full)
train_sds <- apply(X_full, 2, sd)
X_scaled <- scale(X_full, center = train_means, scale = train_sds)
```

```
[16]: #takes ~9min to run
cv_group_lasso_model <- cv.gglasso(
  X_scaled,
  Response,
  group = group,
  loss = "logit",
  pred.loss = "loss",
  nfolds = 10
)

cat("Group Lasso CV Optimized lambda:\n")
print(round(cv_group_lasso_model$lambda.min, 4))
```

```
Group Lasso CV Optimized lambda:
[1] 4e-04
```

```
[17]: group_lasso_model <- gglasso(
  X_scaled,
  Response,
  nlambdas = 100,
  group = group,
  loss = "logit"
)

coefs <- coef(group_lasso_model, s = cv_group_lasso_model$lambda.min)
selected_groups <- unique(group[which(coefs[-1] != 0)]) # exclude intercept

#map group numbers back to variable names
all_vars <- c(factor_vars, numeric_vars)
selected_vars <- all_vars[selected_groups]
cat("Selected variables (entire groups):\n")
print(selected_vars)
```

Selected variables (entire groups):

```
[1] "Race"
[2] "Age_at_Release"
[3] "Residence_PUMA"
[4] "Gang_Affiliated"
[5] "Supervision_Risk_Score_First"
[6] "Supervision_Level_First"
[7] "Education_Level"
[8] "Dependents"
[9] "Prison_Offense"
[10] "Prison_Years"
[11] "Prior_Arrest_Episodes_Felony"
[12] "Prior_Arrest_Episodes_Misd"
[13] "Prior_Arrest_Episodes_Violent"
[14] "Prior_Arrest_Episodes_Property"
[15] "Prior_Arrest_Episodes_Drug"
[16] "Prior_Arrest_Episodes_PPViolationCharges"
[17] "Prior_Arrest_Episodes_DVCharges"
[18] "Prior_Arrest_Episodes_GunCharges"
[19] "Prior_Conviction_Episodes_Felony"
[20] "Prior_Conviction_Episodes_Misd"
[21] "Prior_Conviction_Episodes_Viol"
[22] "Prior_Conviction_Episodes_Prop"
[23] "Prior_Conviction_Episodes_Drug"
[24] "Prior_Conviction_Episodes_PPViolationCharges"
[25] "Prior_Conviction_Episodes_DomesticViolenceCharges"
[26] "Prior_Conviction_Episodes_GunCharges"
[27] "Prior_Revocations_Parole"
[28] "Prior_Revocations_Probation"
[29] "Condition_MH_SA"
[30] "Condition_Cog_Ed"
```

```

[31] "Condition_Other"
[32] "Violations_ElectronicMonitoring"
[33] "Violations_Instruction"
[34] "Violations_FailToReport"
[35] "Violations_MoveWithoutPermission"
[36] "Delinquency_Reports"
[37] "Program_Attendances"
[38] "Program_UnexcusedAbsences"
[39] "Residence_Changes"
[40] "Avg_Days_per_DrugTest"
[41] "DrugTests_THC_Positive"
[42] "DrugTests_Cocaine_Positive"
[43] "DrugTests_Meth_Positive"
[44] "DrugTests_Other_Positive"
[45] "Percent_Days_Employed"
[46] "Jobs_Per_Year"
[47] "Employment_Exempt"

```

```

[18]: #remove any variables not in training data (just in case)
selected_vars <- intersect(selected_vars, names(train_data))

#formula
formula_str <- paste("Recidivism_Within_2years ~", paste(selected_vars,
  ↪collapse = " + "))
selected_formula <- as.formula(formula_str)

#fit model
selected_logit_model <- glm(selected_formula, data = train_data, family =
  ↪"binomial")
summary(selected_logit_model)

```

Call:

```
glm(formula = selected_formula, family = "binomial", data = train_data)
```

Coefficients:

	Estimate	Std. Error
(Intercept)	4.769e-01	2.093e-01
RaceCaucasian	1.420e-01	3.761e-02
Age_at_Release23-27	-2.824e-01	6.948e-02
Age_at_Release28-32	-6.496e-01	7.574e-02
Age_at_Release33-37	-8.852e-01	8.145e-02
Age_at_Release38-42	-1.043e+00	8.904e-02
Age_at_Release43-47	-1.152e+00	9.390e-02
Age_at_Release48 or older	-1.429e+00	9.482e-02
Residence_PUMA	6.928e-03	2.304e-03
Gang_Affiliated1	7.153e-01	5.004e-02
Gang_AffiliatedUnknown	-3.870e-01	5.310e-02

Supervision_Risk_Score_First	2.412e-02	1.027e-02
Supervision_Level_FirstSpecialized	5.831e-02	4.862e-02
Supervision_Level_FirstStandard	-1.962e-01	4.417e-02
Supervision_Level_FirstUnknown	3.929e-02	7.301e-02
Education_LevelHigh School Diploma	6.811e-02	4.754e-02
Education_LevelLess than HS diploma	-1.149e-01	4.952e-02
Dependents1	6.137e-02	4.561e-02
Dependents2	7.228e-02	4.851e-02
Dependents3 or more	1.644e-02	4.331e-02
Prison_OffenseOther	1.897e-01	6.470e-02
Prison_OffenseProperty	1.621e-01	5.499e-02
Prison_OffenseViolent/Non-Sex	1.071e-01	6.497e-02
Prison_OffenseViolent/Sex	-9.808e-02	1.245e-01
Prison_OffenseUnknown	1.720e-01	5.869e-02
Prison_YearsGreater than 2 to 3 years	-2.849e-01	4.988e-02
Prison_YearsLess than 1 year	2.027e-01	4.535e-02
Prison_YearsMore than 3 years	-2.462e-01	5.155e-02
Prior_Arrest_Episodes_Felony1	-1.020e+00	1.860e-01
Prior_Arrest_Episodes_Felony10 or more	-2.047e-01	2.080e-01
Prior_Arrest_Episodes_Felony2	-7.492e-01	1.862e-01
Prior_Arrest_Episodes_Felony3	-7.365e-01	1.887e-01
Prior_Arrest_Episodes_Felony4	-5.452e-01	1.913e-01
Prior_Arrest_Episodes_Felony5	-4.809e-01	1.945e-01
Prior_Arrest_Episodes_Felony6	-4.744e-01	1.977e-01
Prior_Arrest_Episodes_Felony7	-6.150e-01	2.011e-01
Prior_Arrest_Episodes_Felony8	-4.355e-01	2.057e-01
Prior_Arrest_Episodes_Felony9	-3.183e-01	2.103e-01
Prior_Arrest_Episodes_Misd1	-4.636e-03	6.559e-02
Prior_Arrest_Episodes_Misd2	1.607e-02	7.184e-02
Prior_Arrest_Episodes_Misd3	5.826e-03	7.936e-02
Prior_Arrest_Episodes_Misd4	1.451e-01	8.471e-02
Prior_Arrest_Episodes_Misd5	3.147e-02	9.159e-02
Prior_Arrest_Episodes_Misd6 or more	2.259e-01	8.772e-02
Prior_Arrest_Episodes_Violent1	8.886e-02	4.454e-02
Prior_Arrest_Episodes_Violent2	1.350e-01	5.974e-02
Prior_Arrest_Episodes_Violent3 or more	8.103e-02	6.875e-02
Prior_Arrest_Episodes_Property1	8.160e-02	5.459e-02
Prior_Arrest_Episodes_Property2	1.129e-01	6.586e-02
Prior_Arrest_Episodes_Property3	1.668e-01	7.865e-02
Prior_Arrest_Episodes_Property4	1.792e-01	9.079e-02
Prior_Arrest_Episodes_Property5 or more	3.903e-01	9.482e-02
Prior_Arrest_Episodes_Drug1	3.083e-02	5.203e-02
Prior_Arrest_Episodes_Drug2	7.489e-02	6.329e-02
Prior_Arrest_Episodes_Drug3	6.501e-02	7.417e-02
Prior_Arrest_Episodes_Drug4	9.567e-02	8.692e-02
Prior_Arrest_Episodes_Drug5 or more	4.384e-02	8.824e-02
Prior_Arrest_Episodes_PPViolationCharges1	1.428e-01	5.688e-02
Prior_Arrest_Episodes_PPViolationCharges2	2.853e-01	6.519e-02

Prior_Arrest_Episodes_PPViolationCharges3	2.693e-01	7.281e-02
Prior_Arrest_Episodes_PPViolationCharges4	2.702e-01	8.036e-02
Prior_Arrest_Episodes_PPViolationCharges5 or more	4.784e-01	7.817e-02
Prior_Arrest_Episodes_DVCharges1	1.077e-01	5.405e-02
Prior_Arrest_Episodes_GunCharges1	-4.552e-02	4.507e-02
Prior_Conviction_Episodes_Felony1	2.032e-02	4.658e-02
Prior_Conviction_Episodes_Felony2	1.842e-02	5.777e-02
Prior_Conviction_Episodes_Felony3 or more	9.657e-02	6.563e-02
Prior_Conviction_Episodes_Misd1	1.792e-01	5.370e-02
Prior_Conviction_Episodes_Misd2	2.243e-01	6.391e-02
Prior_Conviction_Episodes_Misd3	2.305e-01	7.486e-02
Prior_Conviction_Episodes_Misd4 or more	2.993e-01	7.624e-02
Prior_Conviction_Episodes_Viol1	2.127e-02	4.723e-02
Prior_Conviction_Episodes_Prop1	-1.377e-02	5.051e-02
Prior_Conviction_Episodes_Prop2	-7.953e-02	6.809e-02
Prior_Conviction_Episodes_Prop3 or more	-1.383e-02	8.037e-02
Prior_Conviction_Episodes_Drug1	-1.039e-03	4.988e-02
Prior_Conviction_Episodes_Drug2 or more	-5.190e-02	6.386e-02
Prior_Conviction_Episodes_PPViolationCharges1	-1.404e-01	4.430e-02
Prior_Conviction_Episodes_DomesticViolenceCharges1	1.128e-02	7.146e-02
Prior_Conviction_Episodes_GunCharges1	1.588e-02	5.831e-02
Prior_Revocations_Parole1	3.435e-01	5.747e-02
Prior_Revocations_Probation1	-1.273e-01	4.665e-02
Condition_MH_SA1	2.993e-01	3.844e-02
Condition_Cog_Ed1	1.839e-02	3.629e-02
Condition_Other1	1.072e-01	4.085e-02
Violations_ElectronicMonitoring1	2.202e-01	6.922e-02
Violations_Instruction1	1.238e-01	4.917e-02
Violations_FailToReport1	-3.805e-02	6.542e-02
Violations_MoveWithoutPermission1	-1.638e-01	5.504e-02
Delinquency_Reports1	5.936e-01	8.106e-02
Delinquency_Reports2	-1.144e-01	7.403e-02
Delinquency_Reports3	-4.830e-01	7.760e-02
Delinquency_Reports4 or more	-8.165e-01	5.240e-02
Program_Attendances1	-1.002e-02	8.641e-02
Program_Attendances10 or more	-7.014e-01	6.208e-02
Program_Attendances2	-1.025e-01	9.326e-02
Program_Attendances3	6.945e-02	1.046e-01
Program_Attendances4	-3.101e-01	9.772e-02
Program_Attendances5	-2.599e-01	7.986e-02
Program_Attendances6	-3.851e-01	5.540e-02
Program_Attendances7	-4.040e-01	9.526e-02
Program_Attendances8	-3.211e-01	1.279e-01
Program_Attendances9	-3.423e-01	1.335e-01
Program_UnexcusedAbsences1	2.094e-01	7.124e-02
Program_UnexcusedAbsences2	2.764e-01	8.397e-02
Program_UnexcusedAbsences3 or more	1.766e-01	6.783e-02
Residence_Changes1	1.336e-01	3.966e-02

Residence_Changes2	1.844e-01	5.055e-02
Residence_Changes3 or more	3.547e-01	5.572e-02
Avg_Days_per_DrugTest	5.953e-05	1.664e-04
DrugTests_THC_Positive	8.306e-01	1.486e-01
DrugTests_Cocaine_Positive	6.835e-01	3.069e-01
DrugTests_Meth_Positive	2.286e+00	3.863e-01
DrugTests_Other_Positive	1.121e+00	4.478e-01
Percent_Days_Employed	-1.931e+00	5.415e-02
Jobs_Per_Year	4.575e-01	2.779e-02
Employment_Exempt1	-1.185e-01	4.983e-02
	z value	Pr(> z)
(Intercept)	2.279	0.022667 *
RaceCaucasian	3.775	0.000160 ***
Age_at_Release23-27	-4.064	4.82e-05 ***
Age_at_Release28-32	-8.577	< 2e-16 ***
Age_at_Release33-37	-10.868	< 2e-16 ***
Age_at_Release38-42	-11.713	< 2e-16 ***
Age_at_Release43-47	-12.268	< 2e-16 ***
Age_at_Release48 or older	-15.074	< 2e-16 ***
Residence_PUMA	3.007	0.002639 **
Gang_Affiliated1	14.296	< 2e-16 ***
Gang_AffiliatedUnknown	-7.289	3.13e-13 ***
Supervision_Risk_Score_First	2.349	0.018819 *
Supervision_Level_FirstSpecialized	1.199	0.230414
Supervision_Level_FirstStandard	-4.441	8.94e-06 ***
Supervision_Level_FirstUnknown	0.538	0.590479
Education_LevelHigh School Diploma	1.433	0.151978
Education_LevelLess than HS diploma	-2.319	0.020370 *
Dependents1	1.346	0.178424
Dependents2	1.490	0.136249
Dependents3 or more	0.379	0.704321
Prison_OffenseOther	2.932	0.003365 **
Prison_OffenseProperty	2.948	0.003201 **
Prison_OffenseViolent/Non-Sex	1.649	0.099113 .
Prison_OffenseViolent/Sex	-0.788	0.430627
Prison_OffenseUnknown	2.931	0.003376 **
Prison_YearsGreater than 2 to 3 years	-5.712	1.12e-08 ***
Prison_YearsLess than 1 year	4.469	7.84e-06 ***
Prison_YearsMore than 3 years	-4.776	1.79e-06 ***
Prior_Arrest_Episodes_Felony1	-5.484	4.15e-08 ***
Prior_Arrest_Episodes_Felony10 or more	-0.984	0.324952
Prior_Arrest_Episodes_Felony2	-4.024	5.73e-05 ***
Prior_Arrest_Episodes_Felony3	-3.903	9.50e-05 ***
Prior_Arrest_Episodes_Felony4	-2.850	0.004377 **
Prior_Arrest_Episodes_Felony5	-2.473	0.013412 *
Prior_Arrest_Episodes_Felony6	-2.400	0.016391 *
Prior_Arrest_Episodes_Felony7	-3.057	0.002233 **
Prior_Arrest_Episodes_Felony8	-2.117	0.034236 *

Prior_Arrest_Episodes_Felony9	-1.514	0.130139	
Prior_Arrest_Episodes_Misd1	-0.071	0.943654	
Prior_Arrest_Episodes_Misd2	0.224	0.823036	
Prior_Arrest_Episodes_Misd3	0.073	0.941480	
Prior_Arrest_Episodes_Misd4	1.712	0.086832	.
Prior_Arrest_Episodes_Misd5	0.344	0.731181	
Prior_Arrest_Episodes_Misd6 or more	2.575	0.010015	*
Prior_Arrest_Episodes_Violent1	1.995	0.046041	*
Prior_Arrest_Episodes_Violent2	2.260	0.023818	*
Prior_Arrest_Episodes_Violent3 or more	1.179	0.238556	
Prior_Arrest_Episodes_Property1	1.495	0.135014	
Prior_Arrest_Episodes_Property2	1.715	0.086435	.
Prior_Arrest_Episodes_Property3	2.120	0.033974	*
Prior_Arrest_Episodes_Property4	1.973	0.048462	*
Prior_Arrest_Episodes_Property5 or more	4.116	3.86e-05	***
Prior_Arrest_Episodes_Drug1	0.593	0.553430	
Prior_Arrest_Episodes_Drug2	1.183	0.236707	
Prior_Arrest_Episodes_Drug3	0.877	0.380746	
Prior_Arrest_Episodes_Drug4	1.101	0.271047	
Prior_Arrest_Episodes_Drug5 or more	0.497	0.619360	
Prior_Arrest_Episodes_PPViolationCharges1	2.511	0.012049	*
Prior_Arrest_Episodes_PPViolationCharges2	4.376	1.21e-05	***
Prior_Arrest_Episodes_PPViolationCharges3	3.699	0.000216	***
Prior_Arrest_Episodes_PPViolationCharges4	3.362	0.000773	***
Prior_Arrest_Episodes_PPViolationCharges5 or more	6.120	9.35e-10	***
Prior_Arrest_Episodes_DVCharges1	1.992	0.046317	*
Prior_Arrest_Episodes_GunCharges1	-1.010	0.312512	
Prior_Conviction_Episodes_Felony1	0.436	0.662673	
Prior_Conviction_Episodes_Felony2	0.319	0.749801	
Prior_Conviction_Episodes_Felony3 or more	1.471	0.141164	
Prior_Conviction_Episodes_Misd1	3.337	0.000846	***
Prior_Conviction_Episodes_Misd2	3.510	0.000448	***
Prior_Conviction_Episodes_Misd3	3.079	0.002078	**
Prior_Conviction_Episodes_Misd4 or more	3.926	8.63e-05	***
Prior_Conviction_Episodes_Viol1	0.450	0.652524	
Prior_Conviction_Episodes_Prop1	-0.273	0.785107	
Prior_Conviction_Episodes_Prop2	-1.168	0.242821	
Prior_Conviction_Episodes_Prop3 or more	-0.172	0.863422	
Prior_Conviction_Episodes_Drug1	-0.021	0.983378	
Prior_Conviction_Episodes_Drug2 or more	-0.813	0.416407	
Prior_Conviction_Episodes_PPViolationCharges1	-3.168	0.001533	**
Prior_Conviction_Episodes_DomesticViolenceCharges1	0.158	0.874599	
Prior_Conviction_Episodes_GunCharges1	0.272	0.785374	
Prior_Revocations_Parole1	5.976	2.28e-09	***
Prior_Revocations_Probation1	-2.729	0.006357	**
Condition_MH_SA1	7.786	6.90e-15	***
Condition_Cog_Ed1	0.507	0.612293	
Condition_Other1	2.624	0.008697	**

Violations_ElectronicMonitoring1	3.181	0.001466	**
Violations_Instruction1	2.518	0.011792	*
Violations_FailToReport1	-0.582	0.560856	
Violations_MoveWithoutPermission1	-2.976	0.002916	**
Delinquency_Reports1	7.323	2.43e-13	***
Delinquency_Reports2	-1.546	0.122136	
Delinquency_Reports3	-6.225	4.83e-10	***
Delinquency_Reports4 or more	-15.583	< 2e-16	***
Program_Attendances1	-0.116	0.907683	
Program_Attendances10 or more	-11.297	< 2e-16	***
Program_Attendances2	-1.099	0.271870	
Program_Attendances3	0.664	0.506648	
Program_Attendances4	-3.174	0.001505	**
Program_Attendances5	-3.254	0.001138	**
Program_Attendances6	-6.951	3.62e-12	***
Program_Attendances7	-4.241	2.22e-05	***
Program_Attendances8	-2.510	0.012081	*
Program_Attendances9	-2.565	0.010323	*
Program_UnexcusedAbsences1	2.939	0.003288	**
Program_UnexcusedAbsences2	3.291	0.000997	***
Program_UnexcusedAbsences3 or more	2.603	0.009230	**
Residence_Changes1	3.368	0.000757	***
Residence_Changes2	3.648	0.000265	***
Residence_Changes3 or more	6.365	1.96e-10	***
Avg_Days_per_DrugTest	0.358	0.720517	
DrugTests_THC_Positive	5.588	2.29e-08	***
DrugTests_Cocaine_Positive	2.227	0.025937	*
DrugTests_Meth_Positive	5.918	3.26e-09	***
DrugTests_Other_Positive	2.503	0.012323	*
Percent_Days_Employed	-35.652	< 2e-16	***
Jobs_Per_Year	16.463	< 2e-16	***
Employment_Exempt1	-2.378	0.017404	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28605 on 20667 degrees of freedom
 Residual deviance: 22994 on 20551 degrees of freedom
 AIC: 23228

Number of Fisher Scoring iterations: 4

```
[19]: # Predict probabilities on test set
pred_probs <- predict(selected_logit_model, newdata = test_data, type =
  ↪ "response")
```

```

mean(round(pred_probs,3))
# Classify using 0.5 threshold
pred_class <- ifelse(pred_probs > 0.5, 1, 0)
# Confusion matrix
table(Predicted = pred_class, Actual = as.numeric(as.
  ↪character(test_data$Recidivism_Within_2years)))

```

0.475796593768144

	Actual	
Predicted	0	1
0	1995	762
1	725	1685

```

[20]: # Calculate metrics
TN <- 1995
FP <- 725
FN <- 762
TP <- 1685

accuracy <- (TP + TN) / (TP + TN + FP + FN)
sensitivity <- TP / (TP + FN) # Recall, True Positive Rate
specificity <- TN / (TN + FP) # True Negative Rate
precision <- TP / (TP + FP)

cat(sprintf("Accuracy:    %.3f\n", accuracy))
cat(sprintf("Sensitivity: %.3f\n", sensitivity))
cat(sprintf("Specificity: %.3f\n", specificity))
cat(sprintf("Precision:   %.3f\n", precision))

```

Accuracy: 0.712
 Sensitivity: 0.689
 Specificity: 0.733
 Precision: 0.699

```

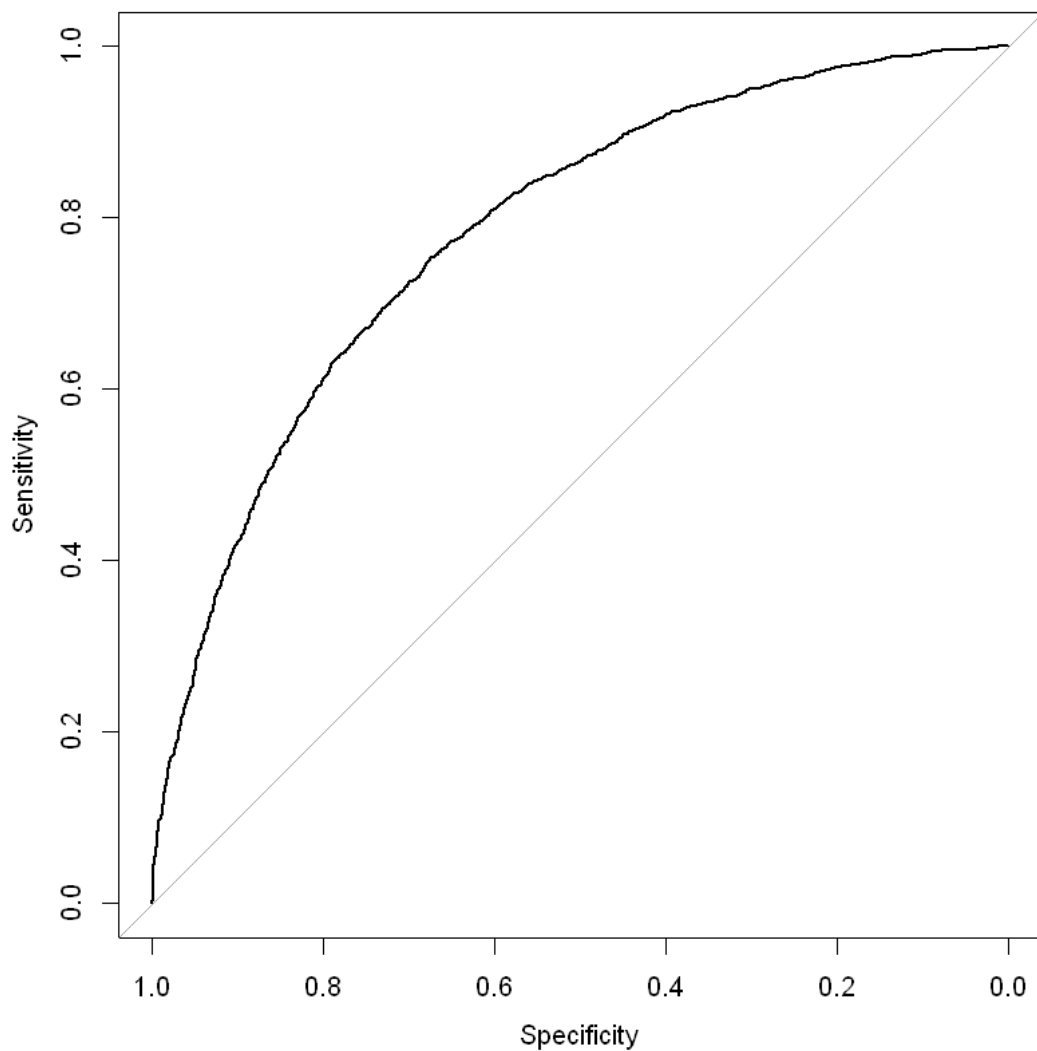
[21]: roc_obj <- roc(test_data$Recidivism_Within_2years, pred_probs)
auc(roc_obj)
plot(roc_obj)

```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

0.784351336570591



1.0.1 We can whittle it down some more

```
[22]: #list of variable name patterns to remove (will match all levels/dummies)
#these variables are removed b/c none of the dummy variables in that particular
      ↪ group had significance (p>0.05)
remove_patterns <- c(
  "^Dependents",
  "^Prior_Arrest_Episodes_Drug",
  "^Prior_Conviction_Episodes_Felony",
  "^Prior_Conviction_Episodes_Prop",
  "^Prior_Conviction_Episodes_Viol",
  "^Prior_Conviction_Episodes_DomesticViolenceCharges",
  "^Prior_Arrest_Episodes_GunCharges",
```

```

    "^Violations_FailToReport",
    "^Condition_Cog_Ed"
)

#remove all variables matching any of the patterns
selected_vars_reduced <- selected_vars
for (pat in remove_patterns) {
  selected_vars_reduced <- selected_vars_reduced[!grepl(pat,
    ↪selected_vars_reduced)]
}

#build new formula
formula_str_reduced <- paste("Recidivism_Within_2years ~",
    ↪paste(selected_vars_reduced, collapse = " + "))
selected_formula_reduced <- as.formula(formula_str_reduced)

#fit model
reduced_logit_model <- glm(selected_formula_reduced, data = train_data, family =
    ↪"binomial")
summary(reduced_logit_model)

#predict probabilities on test set
pred_probs_reduced <- predict(reduced_logit_model, newdata = test_data, type =
    ↪"response")
# Classify using 0.5 threshold
pred_class_reduced <- ifelse(pred_probs_reduced > 0.5, 1, 0)
# Confusion matrix
table(Predicted = pred_class_reduced, Actual = as.numeric(as.
    ↪character(test_data$Recidivism_Within_2years)))

#ROC/AUC
library(pROC)
roc_obj_reduced <- roc(test_data$Recidivism_Within_2years, pred_probs_reduced)
auc(roc_obj_reduced)
plot(roc_obj_reduced)

```

Call:

```
glm(formula = selected_formula_reduced, family = "binomial",
    data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value
(Intercept)	4.957e-01	2.071e-01	2.394
RaceCaucasian	1.415e-01	3.714e-02	3.811
Age_at_Release23-27	-2.760e-01	6.915e-02	-3.991
Age_at_Release28-32	-6.386e-01	7.484e-02	-8.532

Age_at_Release33-37	-8.673e-01	8.004e-02	-10.835
Age_at_Release38-42	-1.025e+00	8.757e-02	-11.705
Age_at_Release43-47	-1.129e+00	9.263e-02	-12.184
Age_at_Release48 or older	-1.404e+00	9.353e-02	-15.009
Residence_PUMA	6.881e-03	2.303e-03	2.988
Gang_Affiliated1	7.155e-01	4.994e-02	14.325
Gang_AffiliatedUnknown	-3.827e-01	5.277e-02	-7.254
Supervision_Risk_Score_First	2.388e-02	1.021e-02	2.338
Supervision_Level_FirstSpecialized	5.711e-02	4.853e-02	1.177
Supervision_Level_FirstStandard	-1.993e-01	4.413e-02	-4.517
Supervision_Level_FirstUnknown	3.942e-02	7.290e-02	0.541
Education_LevelHigh School Diploma	7.570e-02	4.724e-02	1.602
Education_LevelLess than HS diploma	-1.081e-01	4.913e-02	-2.201
Prison_OffenseOther	1.815e-01	6.411e-02	2.831
Prison_OffenseProperty	1.564e-01	5.378e-02	2.908
Prison_OffenseViolent/Non-Sex	1.069e-01	6.334e-02	1.688
Prison_OffenseViolent/Sex	-1.087e-01	1.235e-01	-0.880
Prison_OffenseUnknown	1.690e-01	5.860e-02	2.884
Prison_YearsGreater than 2 to 3 years	-2.872e-01	4.984e-02	-5.763
Prison_YearsLess than 1 year	1.953e-01	4.304e-02	4.537
Prison_YearsMore than 3 years	-2.511e-01	5.119e-02	-4.905
Prior_Arrest_Episodes_Felony1	-1.009e+00	1.850e-01	-5.454
Prior_Arrest_Episodes_Felony10 or more	-1.417e-01	1.977e-01	-0.716
Prior_Arrest_Episodes_Felony2	-7.280e-01	1.842e-01	-3.952
Prior_Arrest_Episodes_Felony3	-7.113e-01	1.860e-01	-3.825
Prior_Arrest_Episodes_Felony4	-5.178e-01	1.879e-01	-2.756
Prior_Arrest_Episodes_Felony5	-4.467e-01	1.902e-01	-2.349
Prior_Arrest_Episodes_Felony6	-4.333e-01	1.926e-01	-2.250
Prior_Arrest_Episodes_Felony7	-5.736e-01	1.955e-01	-2.935
Prior_Arrest_Episodes_Felony8	-3.901e-01	1.989e-01	-1.961
Prior_Arrest_Episodes_Felony9	-2.704e-01	2.030e-01	-1.332
Prior_Arrest_Episodes_Misd1	1.128e-03	6.526e-02	0.017
Prior_Arrest_Episodes_Misd2	2.630e-02	7.099e-02	0.370
Prior_Arrest_Episodes_Misd3	1.857e-02	7.803e-02	0.238
Prior_Arrest_Episodes_Misd4	1.598e-01	8.301e-02	1.925
Prior_Arrest_Episodes_Misd5	4.460e-02	8.956e-02	0.498
Prior_Arrest_Episodes_Misd6 or more	2.445e-01	8.414e-02	2.906
Prior_Arrest_Episodes_Violent1	9.347e-02	4.233e-02	2.208
Prior_Arrest_Episodes_Violent2	1.418e-01	5.536e-02	2.561
Prior_Arrest_Episodes_Violent3 or more	8.947e-02	6.227e-02	1.437
Prior_Arrest_Episodes_Property1	7.215e-02	5.149e-02	1.401
Prior_Arrest_Episodes_Property2	8.898e-02	5.831e-02	1.526
Prior_Arrest_Episodes_Property3	1.337e-01	6.824e-02	1.959
Prior_Arrest_Episodes_Property4	1.447e-01	7.815e-02	1.851
Prior_Arrest_Episodes_Property5 or more	3.769e-01	7.321e-02	5.148
Prior_Arrest_Episodes_PPViolationCharges1	1.435e-01	5.671e-02	2.531
Prior_Arrest_Episodes_PPViolationCharges2	2.864e-01	6.473e-02	4.425
Prior_Arrest_Episodes_PPViolationCharges3	2.667e-01	7.210e-02	3.700

Prior_Arrest_Episodes_PPViolationCharges4	2.682e-01	7.935e-02	3.380
Prior_Arrest_Episodes_PPViolationCharges5 or more	4.783e-01	7.637e-02	6.263
Prior_Arrest_Episodes_DVCharges1	1.169e-01	4.876e-02	2.398
Prior_Conviction_Episodes_Misd1	1.745e-01	5.314e-02	3.283
Prior_Conviction_Episodes_Misd2	2.181e-01	6.266e-02	3.480
Prior_Conviction_Episodes_Misd3	2.221e-01	7.289e-02	3.047
Prior_Conviction_Episodes_Misd4 or more	2.925e-01	7.329e-02	3.991
Prior_Conviction_Episodes_Drug1	3.224e-02	4.332e-02	0.744
Prior_Conviction_Episodes_Drug2 or more	-3.611e-03	5.021e-02	-0.072
Prior_Conviction_Episodes_PPViolationCharges1	-1.307e-01	4.342e-02	-3.011
Prior_Conviction_Episodes_GunCharges1	-1.197e-02	5.045e-02	-0.237
Prior_Revocations_Parole1	3.413e-01	5.732e-02	5.955
Prior_Revocations_Probation1	-1.275e-01	4.658e-02	-2.737
Condition_MH_SA1	3.057e-01	3.779e-02	8.090
Condition_Other1	1.087e-01	4.053e-02	2.683
Violations_ElectronicMonitoring1	2.200e-01	6.914e-02	3.182
Violations_Instruction1	1.200e-01	4.862e-02	2.469
Violations_MoveWithoutPermission1	-1.672e-01	5.478e-02	-3.052
Delinquency_Reports1	5.928e-01	8.096e-02	7.322
Delinquency_Reports2	-1.182e-01	7.386e-02	-1.600
Delinquency_Reports3	-4.836e-01	7.746e-02	-6.243
Delinquency_Reports4 or more	-8.202e-01	5.173e-02	-15.854
Program_Attendances1	-7.256e-03	8.633e-02	-0.084
Program_Attendances10 or more	-6.985e-01	6.195e-02	-11.276
Program_Attendances2	-1.028e-01	9.316e-02	-1.103
Program_Attendances3	6.686e-02	1.045e-01	0.640
Program_Attendances4	-3.046e-01	9.757e-02	-3.122
Program_Attendances5	-2.576e-01	7.979e-02	-3.228
Program_Attendances6	-3.846e-01	5.533e-02	-6.951
Program_Attendances7	-4.032e-01	9.517e-02	-4.237
Program_Attendances8	-3.246e-01	1.278e-01	-2.540
Program_Attendances9	-3.393e-01	1.332e-01	-2.547
Program_UnexcusedAbsences1	2.095e-01	7.118e-02	2.944
Program_UnexcusedAbsences2	2.733e-01	8.389e-02	3.257
Program_UnexcusedAbsences3 or more	1.760e-01	6.773e-02	2.599
Residence_Changes1	1.345e-01	3.962e-02	3.395
Residence_Changes2	1.857e-01	5.051e-02	3.677
Residence_Changes3 or more	3.556e-01	5.565e-02	6.390
Avg_Days_per_DrugTest	6.642e-05	1.661e-04	0.400
DrugTests_THC_Positive	8.277e-01	1.486e-01	5.571
DrugTests_Cocaine_Positive	6.855e-01	3.067e-01	2.235
DrugTests_Meth_Positive	2.271e+00	3.859e-01	5.883
DrugTests_Other_Positive	1.127e+00	4.473e-01	2.520
Percent_Days_Employed	-1.931e+00	5.409e-02	-35.702
Jobs_Per_Year	4.593e-01	2.776e-02	16.545
Employment_Exempt1	-1.189e-01	4.979e-02	-2.388
	Pr(> z)		
(Intercept)	0.016684 *		

RaceCaucasian	0.000138 ***
Age_at_Release23-27	6.57e-05 ***
Age_at_Release28-32	< 2e-16 ***
Age_at_Release33-37	< 2e-16 ***
Age_at_Release38-42	< 2e-16 ***
Age_at_Release43-47	< 2e-16 ***
Age_at_Release48 or older	< 2e-16 ***
Residence_PUMA	0.002807 **
Gang_Affiliated1	< 2e-16 ***
Gang_AffiliatedUnknown	4.06e-13 ***
Supervision_Risk_Score_First	0.019381 *
Supervision_Level_FirstSpecialized	0.239324
Supervision_Level_FirstStandard	6.28e-06 ***
Supervision_Level_FirstUnknown	0.588696
Education_LevelHigh School Diploma	0.109075
Education_LevelLess than HS diploma	0.027710 *
Prison_OffenseOther	0.004638 **
Prison_OffenseProperty	0.003636 **
Prison_OffenseViolent/Non-Sex	0.091457 .
Prison_OffenseViolent/Sex	0.378593
Prison_OffenseUnknown	0.003932 **
Prison_YearsGreater than 2 to 3 years	8.28e-09 ***
Prison_YearsLess than 1 year	5.69e-06 ***
Prison_YearsMore than 3 years	9.32e-07 ***
Prior_Arrest_Episodes_Felony1	4.92e-08 ***
Prior_Arrest_Episodes_Felony10 or more	0.473711
Prior_Arrest_Episodes_Felony2	7.76e-05 ***
Prior_Arrest_Episodes_Felony3	0.000131 ***
Prior_Arrest_Episodes_Felony4	0.005851 **
Prior_Arrest_Episodes_Felony5	0.018834 *
Prior_Arrest_Episodes_Felony6	0.024460 *
Prior_Arrest_Episodes_Felony7	0.003338 **
Prior_Arrest_Episodes_Felony8	0.049866 *
Prior_Arrest_Episodes_Felony9	0.182908
Prior_Arrest_Episodes_Misd1	0.986208
Prior_Arrest_Episodes_Misd2	0.711039
Prior_Arrest_Episodes_Misd3	0.811860
Prior_Arrest_Episodes_Misd4	0.054259 .
Prior_Arrest_Episodes_Misd5	0.618488
Prior_Arrest_Episodes_Misd6 or more	0.003658 **
Prior_Arrest_Episodes_Violent1	0.027221 *
Prior_Arrest_Episodes_Violent2	0.010441 *
Prior_Arrest_Episodes_Violent3 or more	0.150766
Prior_Arrest_Episodes_Property1	0.161130
Prior_Arrest_Episodes_Property2	0.127048
Prior_Arrest_Episodes_Property3	0.050138 .
Prior_Arrest_Episodes_Property4	0.064101 .
Prior_Arrest_Episodes_Property5 or more	2.63e-07 ***

Prior_Arrest_Episodes_PPViolationCharges1	0.011369 *
Prior_Arrest_Episodes_PPViolationCharges2	9.64e-06 ***
Prior_Arrest_Episodes_PPViolationCharges3	0.000216 ***
Prior_Arrest_Episodes_PPViolationCharges4	0.000724 ***
Prior_Arrest_Episodes_PPViolationCharges5 or more	3.78e-10 ***
Prior_Arrest_Episodes_DVCharges1	0.016467 *
Prior_Conviction_Episodes_Misd1	0.001028 **
Prior_Conviction_Episodes_Misd2	0.000501 ***
Prior_Conviction_Episodes_Misd3	0.002309 **
Prior_Conviction_Episodes_Misd4 or more	6.59e-05 ***
Prior_Conviction_Episodes_Drug1	0.456722
Prior_Conviction_Episodes_Drug2 or more	0.942675
Prior_Conviction_Episodes_PPViolationCharges1	0.002606 **
Prior_Conviction_Episodes_GunCharges1	0.812420
Prior_Revocations_Parole1	2.60e-09 ***
Prior_Revocations_Probation1	0.006201 **
Condition_MH_SA1	5.96e-16 ***
Condition_Other1	0.007300 **
Violations_ElectronicMonitoring1	0.001461 **
Violations_Instruction1	0.013548 *
Violations_MoveWithoutPermission1	0.002276 **
Delinquency_Reports1	2.45e-13 ***
Delinquency_Reports2	0.109492
Delinquency_Reports3	4.28e-10 ***
Delinquency_Reports4 or more	< 2e-16 ***
Program_Attendances1	0.933011
Program_Attendances10 or more	< 2e-16 ***
Program_Attendances2	0.269867
Program_Attendances3	0.522301
Program_Attendances4	0.001797 **
Program_Attendances5	0.001246 **
Program_Attendances6	3.62e-12 ***
Program_Attendances7	2.27e-05 ***
Program_Attendances8	0.011072 *
Program_Attendances9	0.010858 *
Program_UnexcusedAbsences1	0.003244 **
Program_UnexcusedAbsences2	0.001124 **
Program_UnexcusedAbsences3 or more	0.009345 **
Residence_Changes1	0.000685 ***
Residence_Changes2	0.000236 ***
Residence_Changes3 or more	1.66e-10 ***
Avg_Days_per_DrugTest	0.689262
DrugTests_THC_Positive	2.53e-08 ***
DrugTests_Cocaine_Positive	0.025403 *
DrugTests_Meth_Positive	4.02e-09 ***
DrugTests_Other_Positive	0.011737 *
Percent_Days_Employed	< 2e-16 ***
Jobs_Per_Year	< 2e-16 ***

Employment_Exempt1 0.016950 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28605 on 20667 degrees of freedom
Residual deviance: 23005 on 20570 degrees of freedom
AIC: 23201

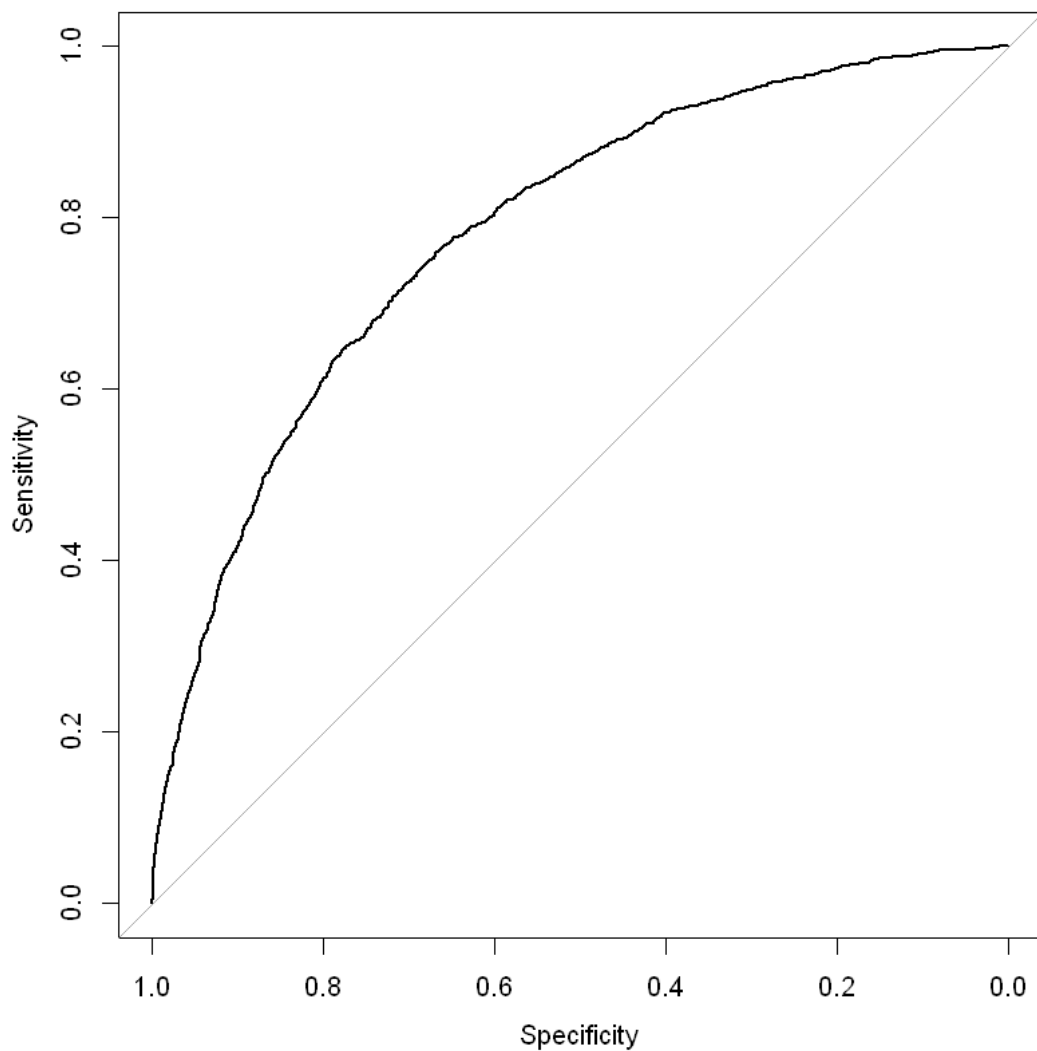
Number of Fisher Scoring iterations: 4

	Actual	
Predicted	0	1
0	1989	764
1	731	1683

Setting levels: control = 0, case = 1

Setting direction: controls < cases

0.784165334503233



1.0.2 Prediction

```
[23]: #predict probabilities on test set
pred_probs <- predict(reduced_logit_model, newdata = test_data, type = "response")
mean(round(pred_probs,3))
#classify using 0.5 threshold
pred_class <- ifelse(pred_probs > 0.5, 1, 0)
#confusion matrix
table(Predicted = pred_class, Actual = as.numeric(as.character(test_data$Recidivism_Within_2years)))
```

0.476035997677569

	Actual	
Predicted	0	1
0	1989	764
1	731	1683

```
[24]: # Calculate metrics
```

```
TN <- 1989
FP <- 731
FN <- 764
TP <- 1683

accuracy <- (TP + TN) / (TP + TN + FP + FN)
sensitivity <- TP / (TP + FN) # Recall, True Positive Rate
specificity <- TN / (TN + FP) # True Negative Rate
precision <- TP / (TP + FP)

cat(sprintf("Accuracy:    %.3f\n", accuracy))
cat(sprintf("Sensitivity: %.3f\n", sensitivity))
cat(sprintf("Specificity: %.3f\n", specificity))
cat(sprintf("Precision:   %.3f\n", precision))
```

```
Accuracy:    0.711
Sensitivity: 0.688
Specificity: 0.731
Precision:   0.697
```

```
[25]: #test for overall regression
```

```
gstat = reduced_logit_model$null.deviance - deviance(reduced_logit_model)
cbind(gstat, 1 - pchisq(gstat, length(coef(reduced_logit_model)) - 1))
```

A matrix: 1×2 of type dbl $\frac{\text{gstat}}{5600.06}$ 0

```
[26]: #Since the p-value is essentially 0 (the number next to the gstat), we reject
      ↪ the null hypothesis that all regression coefficients are zero.
      #Therefore, we can conclude that at least one of the predictors is
      ↪ significantly associated with recidivism.
```

```
[27]: with(summary(reduced_logit_model), c("Residual Deviance" = deviance, "df" = df.
      ↪residual))
pearson_num <- sum(residuals(reduced_logit_model, type = "pearson")^2)
p_value_deviance <- 1 - pchisq(23005, 20570)
p_value_pearson <- 1 - pchisq(pearson_num, 20570)
p_value_deviance
p_value_pearson
```

Residual Deviance	23005.3861442953 df	20570
0		

0.129551698040899

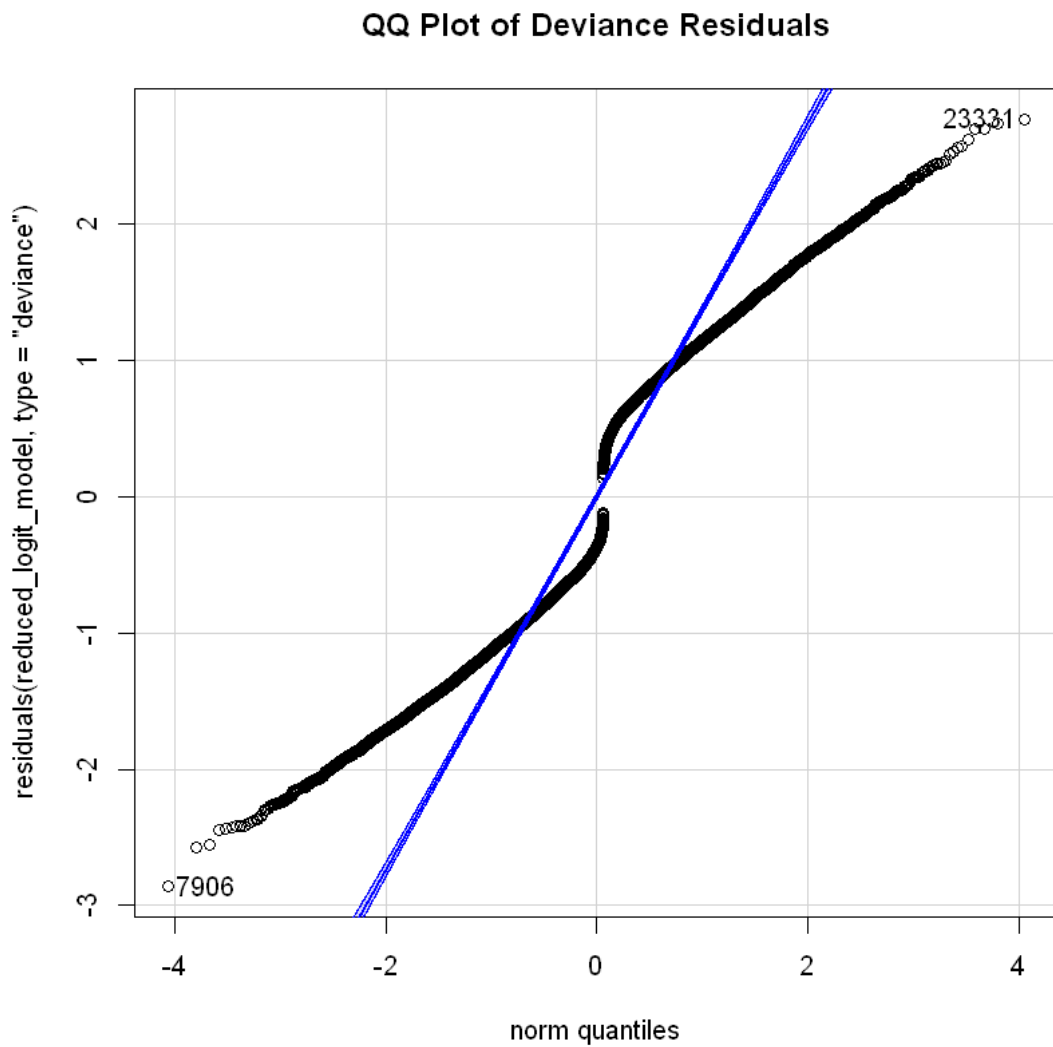
```
[28]: #only the pearson the p-value is over 0.05, so we have mixed results of the  
      ↪null hypothesis that the model fits well.
```

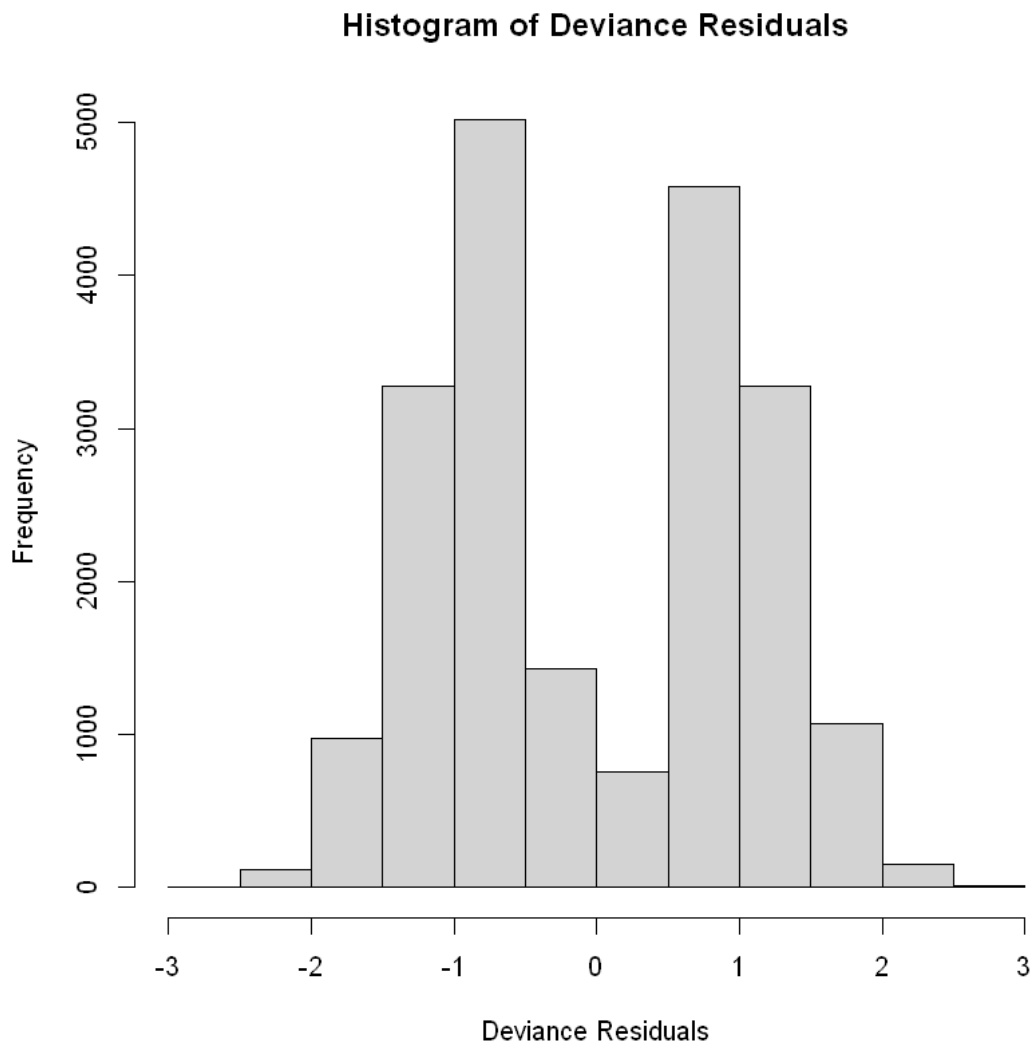
```
[29]: qqPlot(residuals(reduced_logit_model, type = "deviance"), main = "QQ Plot of  
      ↪Deviance Residuals")  
  
hist(residuals(reduced_logit_model, type = "deviance"),  
      main = "Histogram of Deviance Residuals",  
      xlab = "Deviance Residuals",  
      breaks = 20)
```

7906

1690 23331

19187





```
[30]: #normality assumption does NOT hold. residuals look bimodal. Let's investigate_
      ↪numerical variables
num_vars <- c("Supervision_Risk_Score_First", "Avg_Days_per_DrugTest",
              "DrugTests_THC_Positive", "DrugTests_Cocaine_Positive",
              "DrugTests_Meth_Positive", "DrugTests_Other_Positive",
              "Percent_Days_Employed", "Jobs_Per_Year")

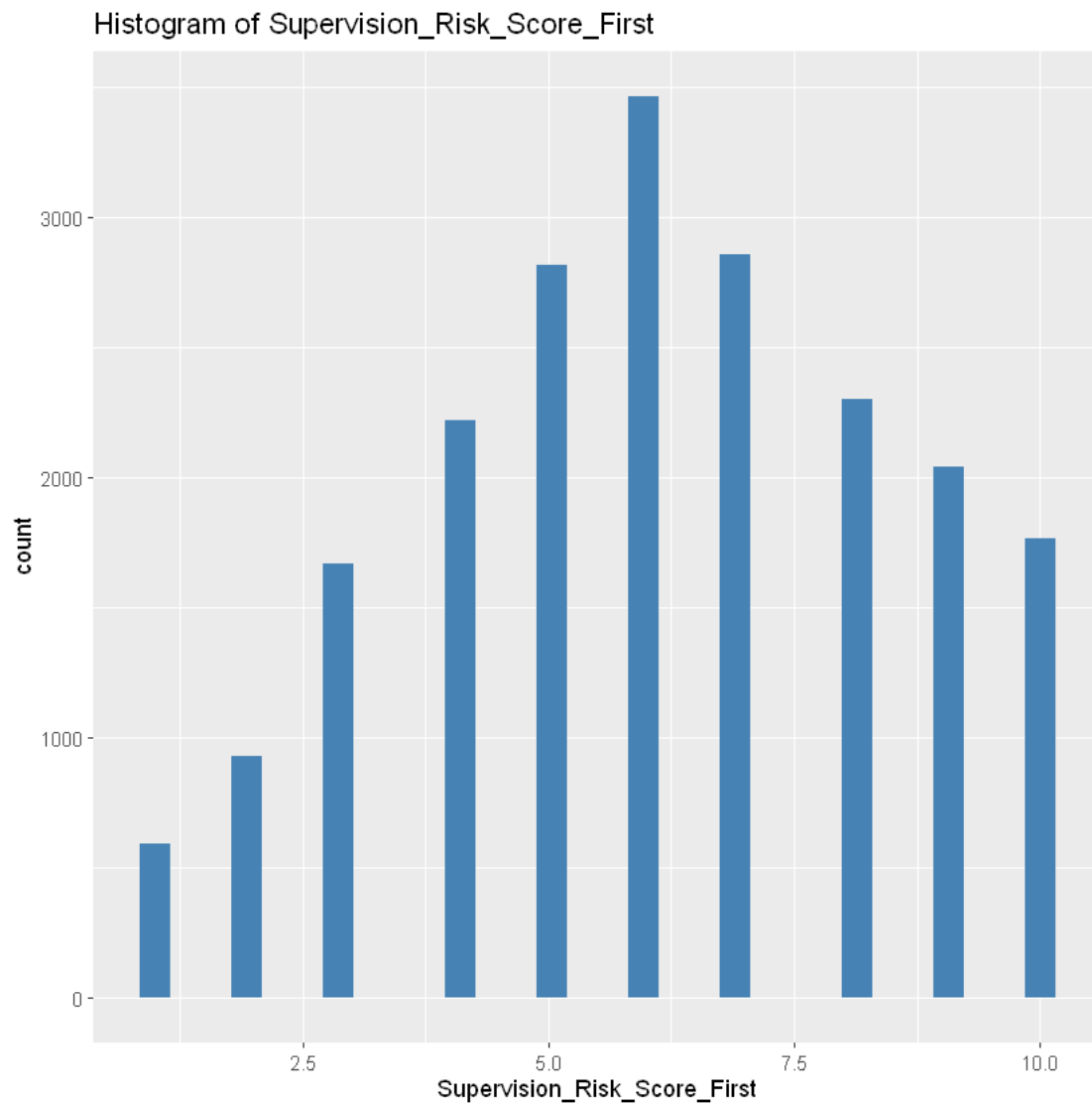
for (var in num_vars) {
  print(ggplot(train_data, aes_string(x = var)) +
        geom_histogram(bins = 30, fill = "steelblue") +
        ggtitle(paste("Histogram of", var)))
}
```

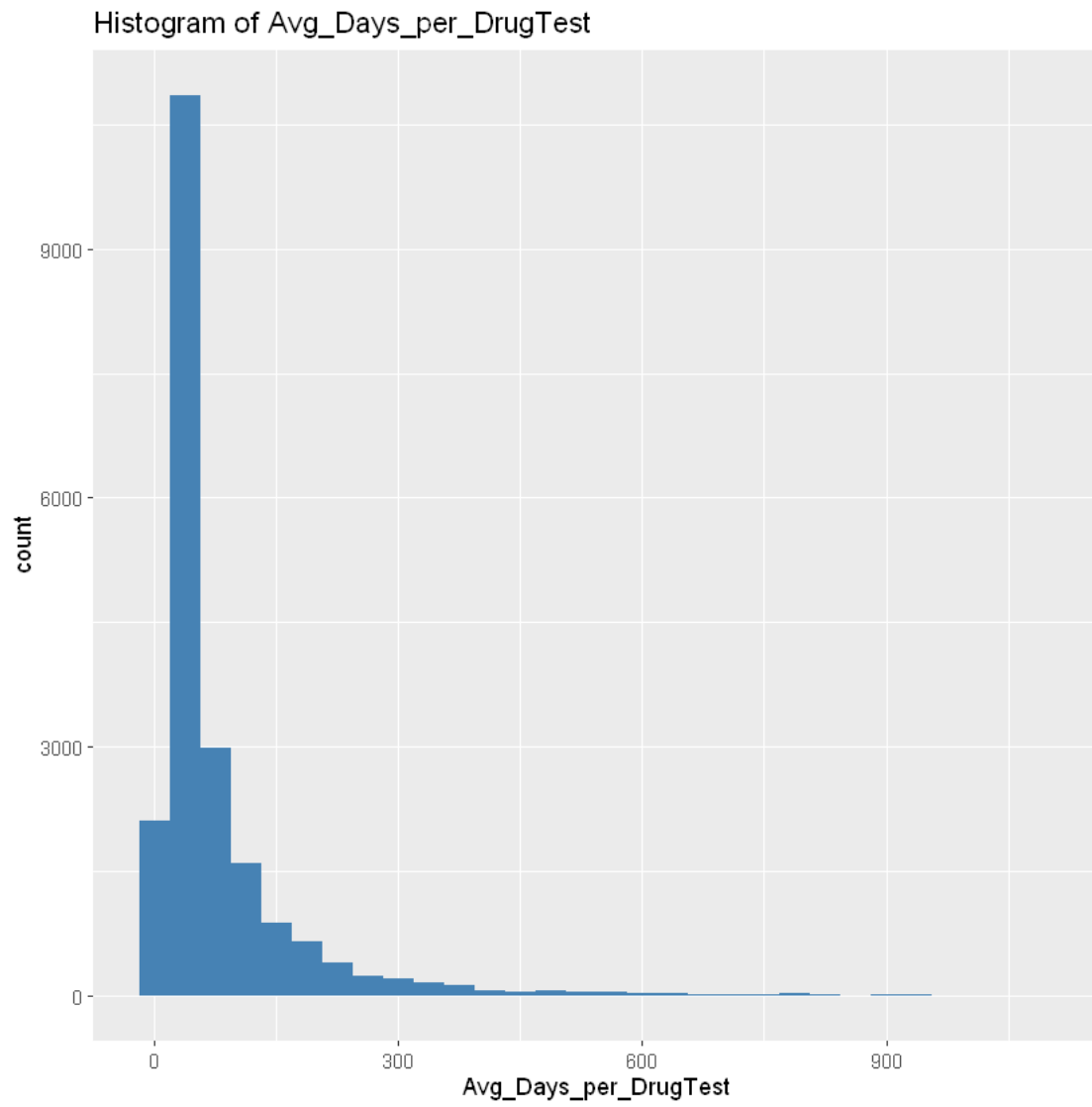
Warning message:

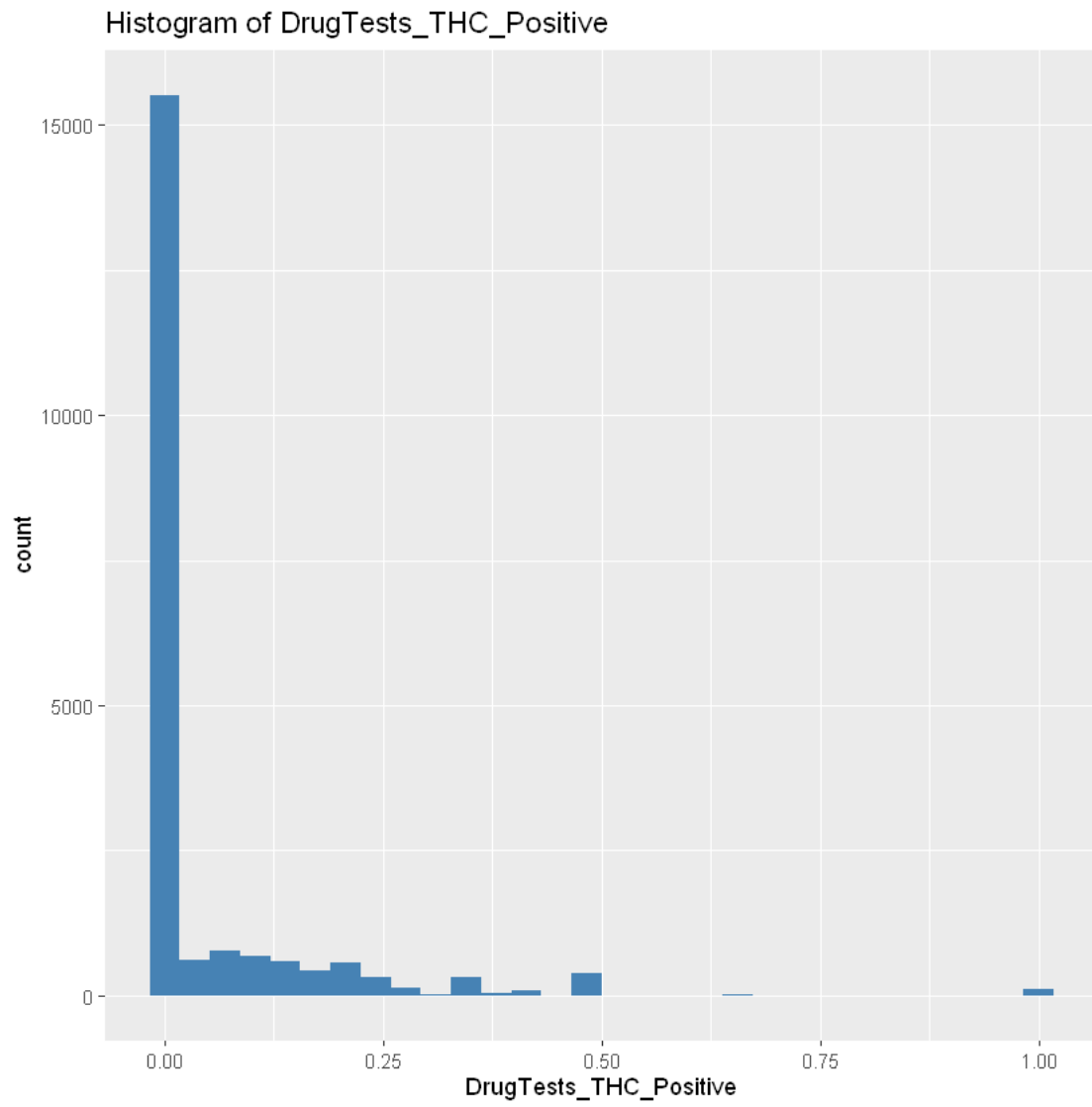
"`aes\_string()` was deprecated in ggplot2 3.0.0.

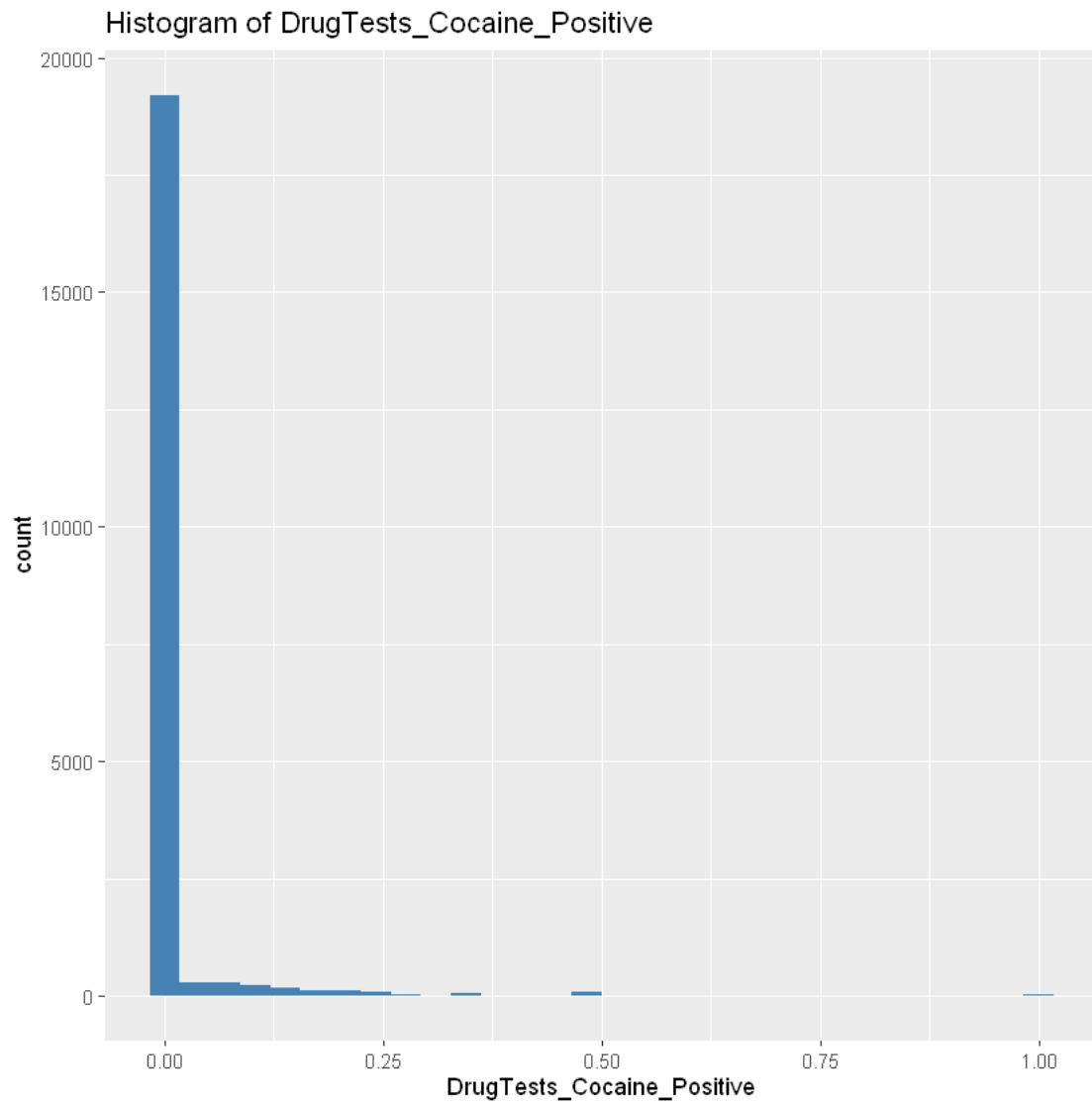
Please use tidy evaluation idioms with `aes()`.

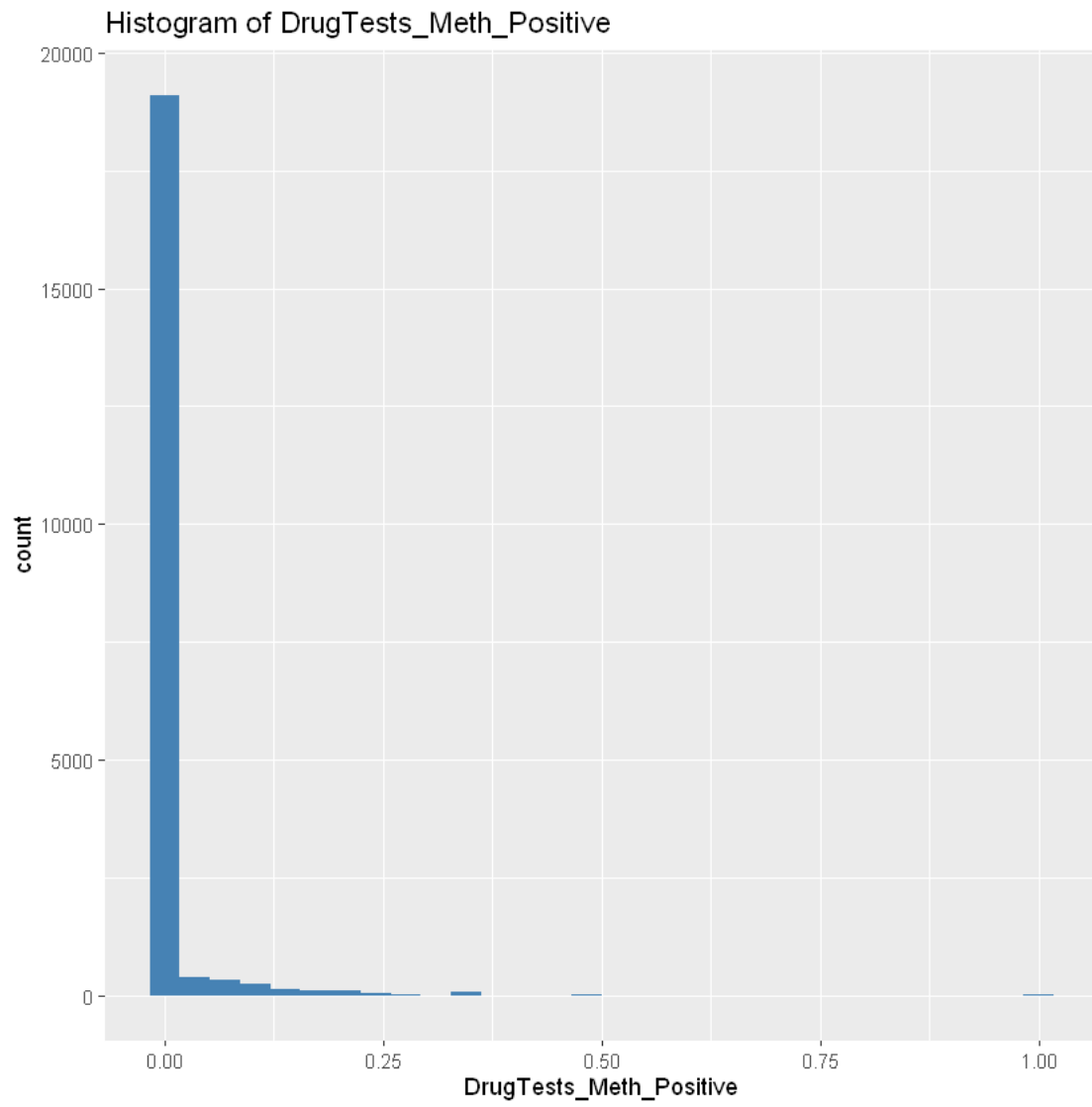
See also `vignette("ggplot2-in-packages")` for more information."

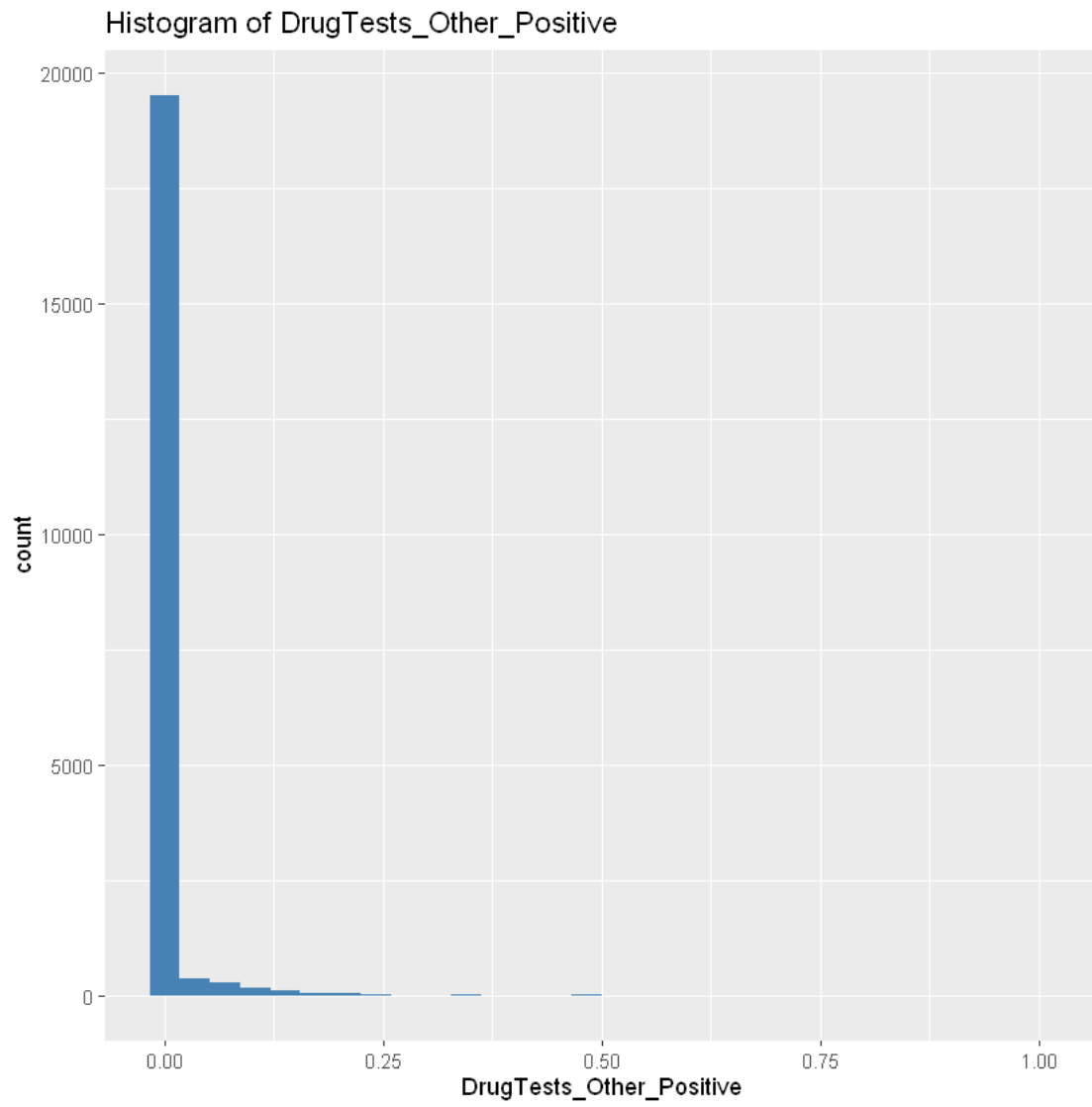


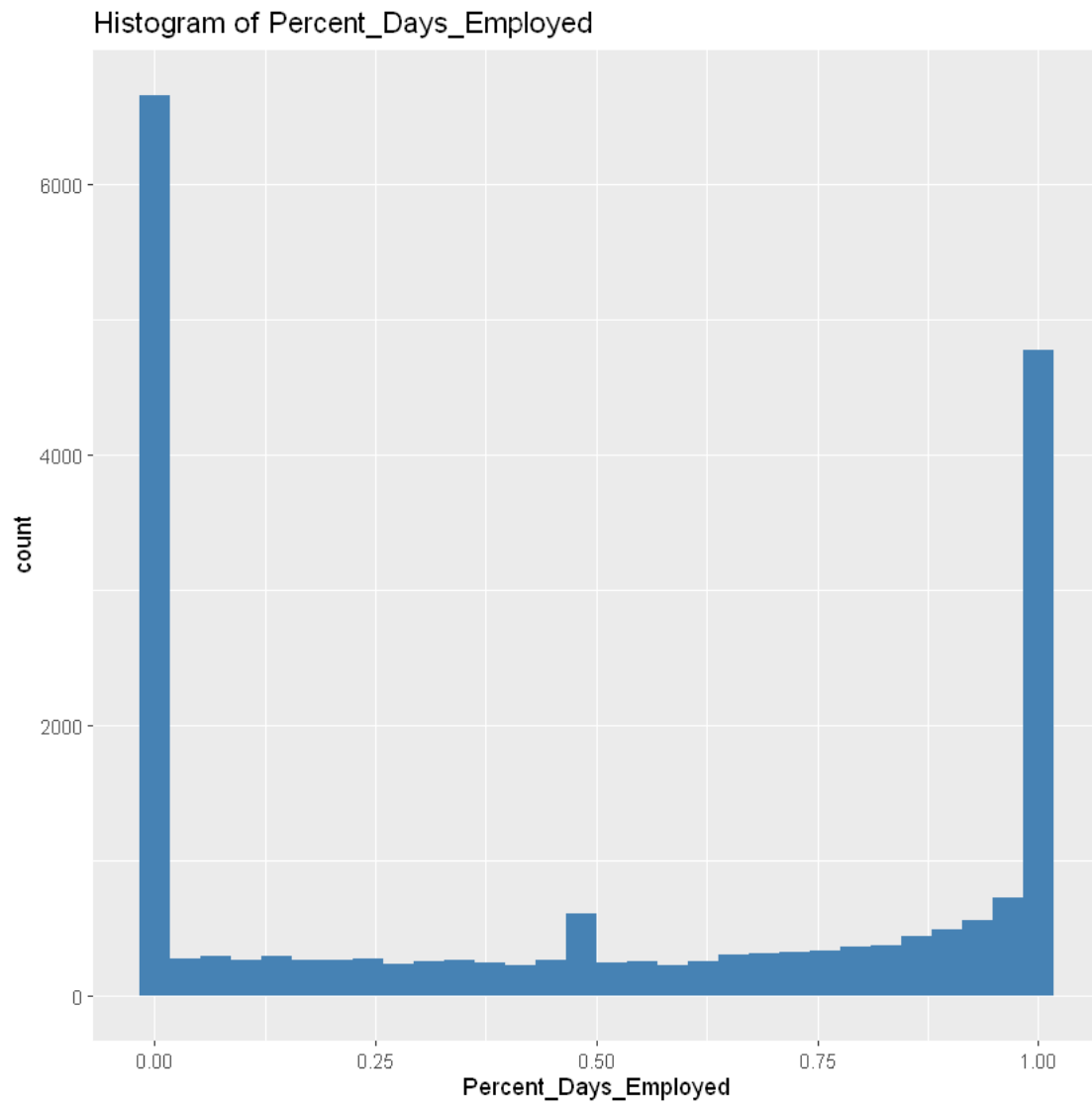


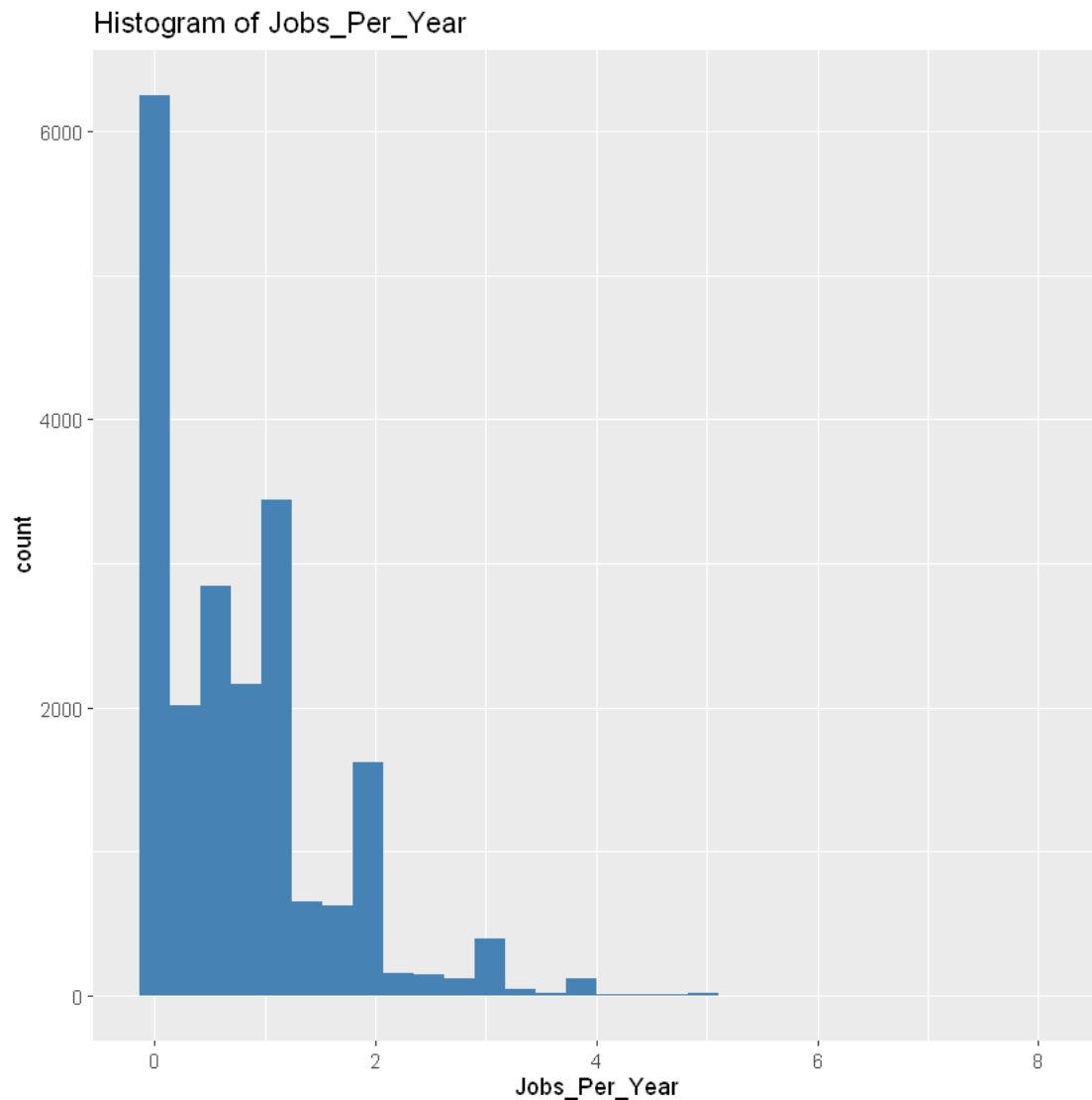












2 Transform Variables

```
[31]: #convert drug test counts to binary (0 = no positive test, 1 = at least one)
train_data$THC_Positive_Flag <-
  factor(ifelse(train_data$DrugTests_THC_Positive > 0, 1, 0), levels = c(0, 1))
train_data$Cocaine_Positive_Flag <-
  factor(ifelse(train_data$DrugTests_Cocaine_Positive > 0, 1, 0), levels =
  c(0, 1))
train_data$Meth_Positive_Flag <-
  factor(ifelse(train_data$DrugTests_Meth_Positive > 0, 1, 0), levels =
  c(0, 1))
```

```

train_data$Other_Positive_Flag <-
  ↪factor(ifelse(train_data$DrugTests_Other_Positive > 0, 1, 0), levels = c(0,
  ↪1))

#handle percent days employed, jobs_per_year, and avg days per drug test

train_data$log_jobs <- log(train_data$Jobs_Per_Year + 1)
train_data$log_Avg_Days_per_DrugTest <- log1p(train_data$Avg_Days_per_DrugTest)
#transform Percent_Days_Employed into three categories
train_data$cat_Percent_Days_Employed <- cut(
  train_data$Percent_Days_Employed,
  breaks = c(0, 0.2, 0.8, 1.0),
  labels = c("Low", "Medium", "High"),
  include.lowest = TRUE
)

#check the distribution
table(train_data$cat_Percent_Days_Employed)

#create dummy variables for logistic regression
employment_dummies <- model.matrix(~ cat_Percent_Days_Employed - 1, data =
  ↪train_data)
employment_dummies <- employment_dummies[, -1] # Remove first column
train_data <- cbind(train_data, employment_dummies)

remove_patterns_new <- c("Avg_Days_per_DrugTest", "Jobs_Per_Year",
  ↪"Percent_Days_Employed",
  "DrugTests_THC_Positive", "DrugTests_Cocaine_Positive",
  "DrugTests_Meth_Positive", "DrugTests_Other_Positive")

#fix the variable selection
selected_vars_reduced_new <- selected_vars_reduced # Fix this line
for (pat in remove_patterns_new) {
  selected_vars_reduced_new <- selected_vars_reduced_new[!grepl(pat,
  ↪selected_vars_reduced_new)]
}

#add new variables with correct dummy variable names
selected_vars_reduced_new <- c(selected_vars_reduced_new,
  "THC_Positive_Flag",
  "Cocaine_Positive_Flag",
  "Meth_Positive_Flag",
  "Other_Positive_Flag",
  "log_Avg_Days_per_DrugTest",
  "log_jobs",
  "cat_Percent_Days_EmployedMedium",
  "cat_Percent_Days_EmployedHigh")

```

Low	Medium	High
8139	5052	7477

```
[32]: formula_str_reduced_new <- paste("Recidivism_Within_2years ~",
  ↪paste(selected_vars_reduced_new, collapse = " + "))
selected_formula_reduced_new <- as.formula(formula_str_reduced_new)

reduced_logit_model_new <- glm(selected_formula_reduced_new, data = train_data,
  ↪family = "binomial")
summary(reduced_logit_model_new)
```

Call:

```
glm(formula = selected_formula_reduced_new, family = "binomial",
  data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value
(Intercept)	0.490314	0.226036	2.169
RaceCaucasian	0.110681	0.037790	2.929
Age_at_Release23-27	-0.292647	0.068836	-4.251
Age_at_Release28-32	-0.663447	0.074479	-8.908
Age_at_Release33-37	-0.891576	0.079715	-11.185
Age_at_Release38-42	-1.043433	0.087221	-11.963
Age_at_Release43-47	-1.145448	0.092304	-12.409
Age_at_Release48 or older	-1.412167	0.093195	-15.153
Residence_PUMA	0.006695	0.002292	2.921
Gang_Affiliated1	0.735311	0.049650	14.810
Gang_AffiliatedUnknown	-0.410017	0.052720	-7.777
Supervision_Risk_Score_First	0.032980	0.010180	3.240
Supervision_Level_FirstSpecialized	0.078463	0.048225	1.627
Supervision_Level_FirstStandard	-0.195345	0.043982	-4.442
Supervision_Level_FirstUnknown	0.036781	0.072505	0.507
Education_LevelHigh School Diploma	0.086016	0.047045	1.828
Education_LevelLess than HS diploma	-0.084549	0.048906	-1.729
Prison_OffenseOther	0.176374	0.063760	2.766
Prison_OffenseProperty	0.158930	0.053478	2.972
Prison_OffenseViolent/Non-Sex	0.110872	0.063049	1.759
Prison_OffenseViolent/Sex	-0.140302	0.123617	-1.135
Prison_OffenseUnknown	0.183107	0.058375	3.137
Prison_YearsGreater than 2 to 3 years	-0.301939	0.049627	-6.084
Prison_YearsLess than 1 year	0.216830	0.042857	5.059
Prison_YearsMore than 3 years	-0.278330	0.051000	-5.457
Prior_Arrest_Episodes_Felony1	-0.990248	0.184862	-5.357
Prior_Arrest_Episodes_Felony10 or more	-0.142850	0.197505	-0.723
Prior_Arrest_Episodes_Felony2	-0.720725	0.184152	-3.914
Prior_Arrest_Episodes_Felony3	-0.723701	0.185871	-3.894

Prior_Arrest_Episodes_Felony4	-0.516797	0.187754	-2.753
Prior_Arrest_Episodes_Felony5	-0.435878	0.189983	-2.294
Prior_Arrest_Episodes_Felony6	-0.433518	0.192359	-2.254
Prior_Arrest_Episodes_Felony7	-0.575358	0.195250	-2.947
Prior_Arrest_Episodes_Felony8	-0.378144	0.198669	-1.903
Prior_Arrest_Episodes_Felony9	-0.268445	0.202727	-1.324
Prior_Arrest_Episodes_Misd1	-0.003247	0.064981	-0.050
Prior_Arrest_Episodes_Misd2	0.032726	0.070744	0.463
Prior_Arrest_Episodes_Misd3	0.017756	0.077709	0.228
Prior_Arrest_Episodes_Misd4	0.130516	0.082704	1.578
Prior_Arrest_Episodes_Misd5	0.035150	0.089189	0.394
Prior_Arrest_Episodes_Misd6 or more	0.237215	0.083822	2.830
Prior_Arrest_Episodes_Violent1	0.098580	0.042111	2.341
Prior_Arrest_Episodes_Violent2	0.146384	0.055072	2.658
Prior_Arrest_Episodes_Violent3 or more	0.105824	0.061899	1.710
Prior_Arrest_Episodes_Property1	0.087469	0.051281	1.706
Prior_Arrest_Episodes_Property2	0.092386	0.058071	1.591
Prior_Arrest_Episodes_Property3	0.151560	0.067982	2.229
Prior_Arrest_Episodes_Property4	0.162738	0.077727	2.094
Prior_Arrest_Episodes_Property5 or more	0.384785	0.072876	5.280
Prior_Arrest_Episodes_PPViolationCharges1	0.152482	0.056524	2.698
Prior_Arrest_Episodes_PPViolationCharges2	0.301542	0.064445	4.679
Prior_Arrest_Episodes_PPViolationCharges3	0.288147	0.071745	4.016
Prior_Arrest_Episodes_PPViolationCharges4	0.291234	0.078977	3.688
Prior_Arrest_Episodes_PPViolationCharges5 or more	0.503150	0.076033	6.618
Prior_Arrest_Episodes_DVCharges1	0.101579	0.048472	2.096
Prior_Conviction_Episodes_Misd1	0.177242	0.052940	3.348
Prior_Conviction_Episodes_Misd2	0.228182	0.062360	3.659
Prior_Conviction_Episodes_Misd3	0.228496	0.072565	3.149
Prior_Conviction_Episodes_Misd4 or more	0.300578	0.072938	4.121
Prior_Conviction_Episodes_Drug1	0.031266	0.043091	0.726
Prior_Conviction_Episodes_Drug2 or more	-0.006001	0.049936	-0.120
Prior_Conviction_Episodes_PPViolationCharges1	-0.140057	0.043173	-3.244
Prior_Conviction_Episodes_GunCharges1	-0.014568	0.050139	-0.291
Prior_Revocations_Parole1	0.338018	0.057029	5.927
Prior_Revocations_Probation1	-0.136981	0.046393	-2.953
Condition_MH_SA1	0.315876	0.037652	8.389
Condition_Other1	0.100987	0.040326	2.504
Violations_ElectronicMonitoring1	0.207086	0.068866	3.007
Violations_Instruction1	0.142075	0.048287	2.942
Violations_MoveWithoutPermission1	-0.152620	0.054459	-2.802
Delinquency_Reports1	0.596415	0.080454	7.413
Delinquency_Reports2	-0.079135	0.073426	-1.078
Delinquency_Reports3	-0.447068	0.077113	-5.798
Delinquency_Reports4 or more	-0.747025	0.052035	-14.356
Program_Attendances1	0.021660	0.085554	0.253
Program_Attendances10 or more	-0.737422	0.063506	-11.612
Program_Attendances2	-0.093186	0.092881	-1.003

Program_Attendances3	0.067641	0.103975	0.651
Program_Attendances4	-0.310060	0.097350	-3.185
Program_Attendances5	-0.281209	0.079864	-3.521
Program_Attendances6	-0.416771	0.055597	-7.496
Program_Attendances7	-0.439653	0.095188	-4.619
Program_Attendances8	-0.356461	0.127653	-2.792
Program_Attendances9	-0.326935	0.132956	-2.459
Program_UnexcusedAbsences1	0.246985	0.070906	3.483
Program_UnexcusedAbsences2	0.345126	0.083365	4.140
Program_UnexcusedAbsences3 or more	0.251109	0.067156	3.739
Residence_Changes1	0.124403	0.039424	3.156
Residence_Changes2	0.173498	0.050306	3.449
Residence_Changes3 or more	0.335008	0.055477	6.039
Employment_Exempt1	-0.120965	0.049638	-2.437
THC_Positive_Flag1	-0.045786	0.043665	-1.049
Cocaine_Positive_Flag1	0.013922	0.066331	0.210
Meth_Positive_Flag1	0.308140	0.067917	4.537
Other_Positive_Flag1	-0.117309	0.073152	-1.604
log_Avg_Days_per_DrugTest	-0.032250	0.020539	-1.570
log_jobs	0.937267	0.061192	15.317
cat_Percent_Days_EmployedMedium	-0.867843	0.051235	-16.938
cat_Percent_Days_EmployedHigh	-1.912608	0.058259	-32.830
	Pr(> z)		
(Intercept)	0.030069	*	
RaceCaucasian	0.003402	**	
Age_at_Release23-27	2.12e-05	***	
Age_at_Release28-32	< 2e-16	***	
Age_at_Release33-37	< 2e-16	***	
Age_at_Release38-42	< 2e-16	***	
Age_at_Release43-47	< 2e-16	***	
Age_at_Release48 or older	< 2e-16	***	
Residence_PUMA	0.003486	**	
Gang_Affiliated1	< 2e-16	***	
Gang_AffiliatedUnknown	7.41e-15	***	
Supervision_Risk_Score_First	0.001196	**	
Supervision_Level_FirstSpecialized	0.103732		
Supervision_Level_FirstStandard	8.93e-06	***	
Supervision_Level_FirstUnknown	0.611949		
Education_LevelHigh School Diploma	0.067492	.	
Education_LevelLess than HS diploma	0.083843	.	
Prison_OffenseOther	0.005671	**	
Prison_OffenseProperty	0.002960	**	
Prison_OffenseViolent/Non-Sex	0.078662	.	
Prison_OffenseViolent/Sex	0.256386		
Prison_OffenseUnknown	0.001708	**	
Prison_YearsGreater than 2 to 3 years	1.17e-09	***	
Prison_YearsLess than 1 year	4.21e-07	***	
Prison_YearsMore than 3 years	4.83e-08	***	

Prior_Arrest_Episodes_Felony1	8.48e-08 ***
Prior_Arrest_Episodes_Felony10 or more	0.469511
Prior_Arrest_Episodes_Felony2	9.09e-05 ***
Prior_Arrest_Episodes_Felony3	9.88e-05 ***
Prior_Arrest_Episodes_Felony4	0.005914 **
Prior_Arrest_Episodes_Felony5	0.021774 *
Prior_Arrest_Episodes_Felony6	0.024215 *
Prior_Arrest_Episodes_Felony7	0.003211 **
Prior_Arrest_Episodes_Felony8	0.056990 .
Prior_Arrest_Episodes_Felony9	0.185447
Prior_Arrest_Episodes_Misd1	0.960144
Prior_Arrest_Episodes_Misd2	0.643658
Prior_Arrest_Episodes_Misd3	0.819263
Prior_Arrest_Episodes_Misd4	0.114541
Prior_Arrest_Episodes_Misd5	0.693501
Prior_Arrest_Episodes_Misd6 or more	0.004655 **
Prior_Arrest_Episodes_Violent1	0.019236 *
Prior_Arrest_Episodes_Violent2	0.007859 **
Prior_Arrest_Episodes_Violent3 or more	0.087335 .
Prior_Arrest_Episodes_Property1	0.088065 .
Prior_Arrest_Episodes_Property2	0.111631
Prior_Arrest_Episodes_Property3	0.025785 *
Prior_Arrest_Episodes_Property4	0.036285 *
Prior_Arrest_Episodes_Property5 or more	1.29e-07 ***
Prior_Arrest_Episodes_PPViolationCharges1	0.006983 **
Prior_Arrest_Episodes_PPViolationCharges2	2.88e-06 ***
Prior_Arrest_Episodes_PPViolationCharges3	5.91e-05 ***
Prior_Arrest_Episodes_PPViolationCharges4	0.000226 ***
Prior_Arrest_Episodes_PPViolationCharges5 or more	3.65e-11 ***
Prior_Arrest_Episodes_DVCharges1	0.036114 *
Prior_Conviction_Episodes_Misd1	0.000814 ***
Prior_Conviction_Episodes_Misd2	0.000253 ***
Prior_Conviction_Episodes_Misd3	0.001639 **
Prior_Conviction_Episodes_Misd4 or more	3.77e-05 ***
Prior_Conviction_Episodes_Drug1	0.468090
Prior_Conviction_Episodes_Drug2 or more	0.904340
Prior_Conviction_Episodes_PPViolationCharges1	0.001178 **
Prior_Conviction_Episodes_GunCharges1	0.771400
Prior_Revocations_Parole1	3.08e-09 ***
Prior_Revocations_Probation1	0.003151 **
Condition_MH_SA1	< 2e-16 ***
Condition_Other1	0.012271 *
Violations_ElectronicMonitoring1	0.002638 **
Violations_Instruction1	0.003258 **
Violations_MoveWithoutPermission1	0.005071 **
Delinquency_Reports1	1.23e-13 ***
Delinquency_Reports2	0.281144
Delinquency_Reports3	6.73e-09 ***

Delinquency_Reports4 or more	< 2e-16 ***
Program_Attendances1	0.800136
Program_Attendances10 or more	< 2e-16 ***
Program_Attendances2	0.315722
Program_Attendances3	0.515338
Program_Attendances4	0.001448 **
Program_Attendances5	0.000430 ***
Program_Attendances6	6.57e-14 ***
Program_Attendances7	3.86e-06 ***
Program_Attendances8	0.005232 **
Program_Attendances9	0.013933 *
Program_UnexcusedAbsences1	0.000495 ***
Program_UnexcusedAbsences2	3.47e-05 ***
Program_UnexcusedAbsences3 or more	0.000185 ***
Residence_Changes1	0.001602 **
Residence_Changes2	0.000563 ***
Residence_Changes3 or more	1.55e-09 ***
Employment_Exempt1	0.014812 *
THC_Positive_Flag1	0.294374
Cocaine_Positive_Flag1	0.833762
Meth_Positive_Flag1	5.71e-06 ***
Other_Positive_Flag1	0.108794
log_Avg_Days_per_DrugTest	0.116382
log_jobs	< 2e-16 ***
cat_Percent_Days_EmployedMedium	< 2e-16 ***
cat_Percent_Days_EmployedHigh	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28605 on 20667 degrees of freedom

Residual deviance: 23187 on 20569 degrees of freedom

AIC: 23385

Number of Fisher Scoring iterations: 4

```
[33]: # Apply the same transformations to test_data that we did to train_data
set.seed(100)
#convert drug test counts to binary flags
test_data$THC_Positive_Flag <-
  factor(ifelse(test_data$DrugTests_THC_Positive > 0, 1, 0), levels = c(0, 1))
test_data$Cocaine_Positive_Flag <-
  factor(ifelse(test_data$DrugTests_Cocaine_Positive > 0, 1, 0), levels = c(0,
  1))
```

```

test_data$Meth_Positive_Flag <-
  ↪factor(ifelse(test_data$DrugTests_Meth_Positive > 0, 1, 0), levels = c(0, 1))
test_data$Other_Positive_Flag <-
  ↪factor(ifelse(test_data$DrugTests_Other_Positive > 0, 1, 0), levels = c(0,
  ↪1))

#create log transformations
test_data$log_jobs <- log(test_data$Jobs_Per_Year + 1)
test_data$log_Avg_Days_per_DrugTest <- log1p(test_data$Avg_Days_per_DrugTest)

#transform employment into categories
test_data$cat_Percent_Days_Employed <- cut(
  test_data$Percent_Days_Employed,
  breaks = c(0, 0.2, 0.8, 1.0),
  labels = c("Low", "Medium", "High"),
  include.lowest = TRUE
)

#remove original variables
test_data$Jobs_Per_Year <- NULL
test_data$Avg_Days_per_DrugTest <- NULL
test_data$Percent_Days_Employed <- NULL
test_data$DrugTests_Other_Positive <- NULL
test_data$DrugTests_THC_Positive <- NULL
test_data$DrugTests_Cocaine_Positive <- NULL
test_data$DrugTests_Meth_Positive <- NULL

#create dummy variables for employment categories
employment_dummies_test <- model.matrix(~ cat_Percent_Days_Employed - 1, data =
  ↪test_data)
employment_dummies_test <- employment_dummies_test[, -1] # Remove first column
test_data <- cbind(test_data, employment_dummies_test)

#predict probabilities on test set
pred_probs_new <- predict(reduced_logit_model_new, newdata = test_data, type =
  ↪"response")
mean(round(pred_probs_new,3))
#classify using 0.5 threshold
pred_class_new <- ifelse(pred_probs_new > 0.5, 1, 0)
#confusion matrix
table(Predicted = pred_class_new, Actual = as.numeric(as.
  ↪character(test_data$Recidivism_Within_2years)))

roc_obj_new <- roc(test_data$Recidivism_Within_2years, pred_probs_new)
auc(roc_obj_new)
plot(roc_obj_new)

```

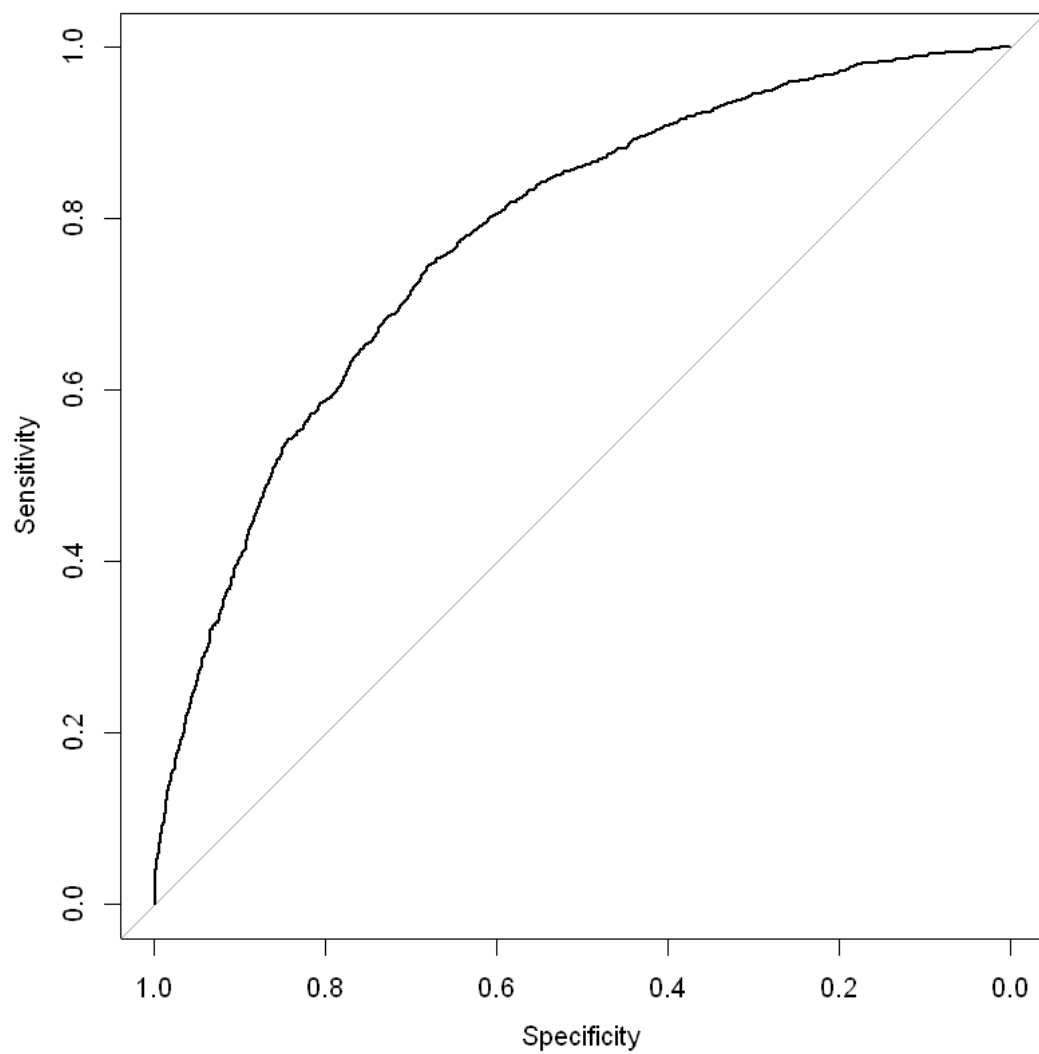
0.477093864911941

	Actual	
Predicted	0	1
0	1950	754
1	770	1693

Setting levels: control = 0, case = 1

Setting direction: controls < cases

0.777305193634462



```
[34]: #calculate metrics
TN <- 1950
FP <- 770
FN <- 754
TP <- 1693

accuracy <- (TP + TN) / (TP + TN + FP + FN)
sensitivity <- TP / (TP + FN) # Recall, True Positive Rate
specificity <- TN / (TN + FP) # True Negative Rate
precision <- TP / (TP + FP)

cat(sprintf("Accuracy:    %.3f\n", accuracy))
cat(sprintf("Sensitivity: %.3f\n", sensitivity))
cat(sprintf("Specificity: %.3f\n", specificity))
cat(sprintf("Precision:   %.3f\n", precision))
```

```
Accuracy:    0.705
Sensitivity: 0.692
Specificity: 0.717
Precision:   0.687
```

```
[35]: #test for overall regression
gstat = reduced_logit_model_new$null.deviance -
  ↪deviance(reduced_logit_model_new)
cbind(gstat, 1 - pchisq(gstat, length(coef(reduced_logit_model_new)) - 1))
```

A matrix: 1×2 of type dbl $\frac{\text{gstat}}{5418.252 \quad 0}$

```
[36]: with(summary(reduced_logit_model_new), c("Residual Deviance" = deviance, "df" =
  ↪df.residual))
pearson_num <- sum(residuals(reduced_logit_model_new, type = "pearson")^2)
p_value_deviance <- 1 - pchisq(23187, 20569)
p_value_pearson <- 1 - pchisq(pearson_num, 20569)
p_value_deviance
p_value_pearson
```

```
Residual Deviance      23187.1948764473 df      20569
0
0.205004183169954
```

```
[37]: qqPlot(residuals(reduced_logit_model_new, type = "deviance"), main = "QQ Plot
  ↪of Deviance Residuals")

hist(residuals(reduced_logit_model_new, type = "deviance"),
      main = "Histogram of Deviance Residuals",
      xlab = "Deviance Residuals",
```

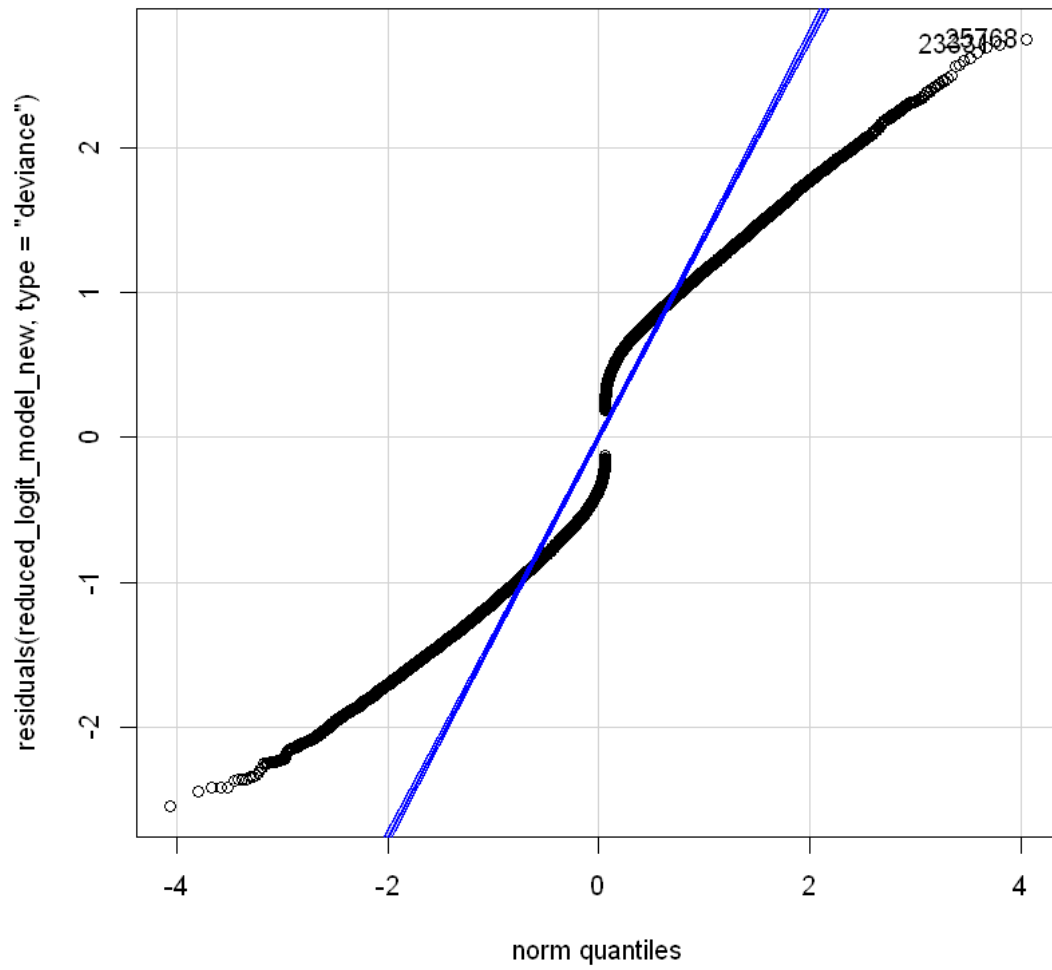
```
breaks = 20)
```

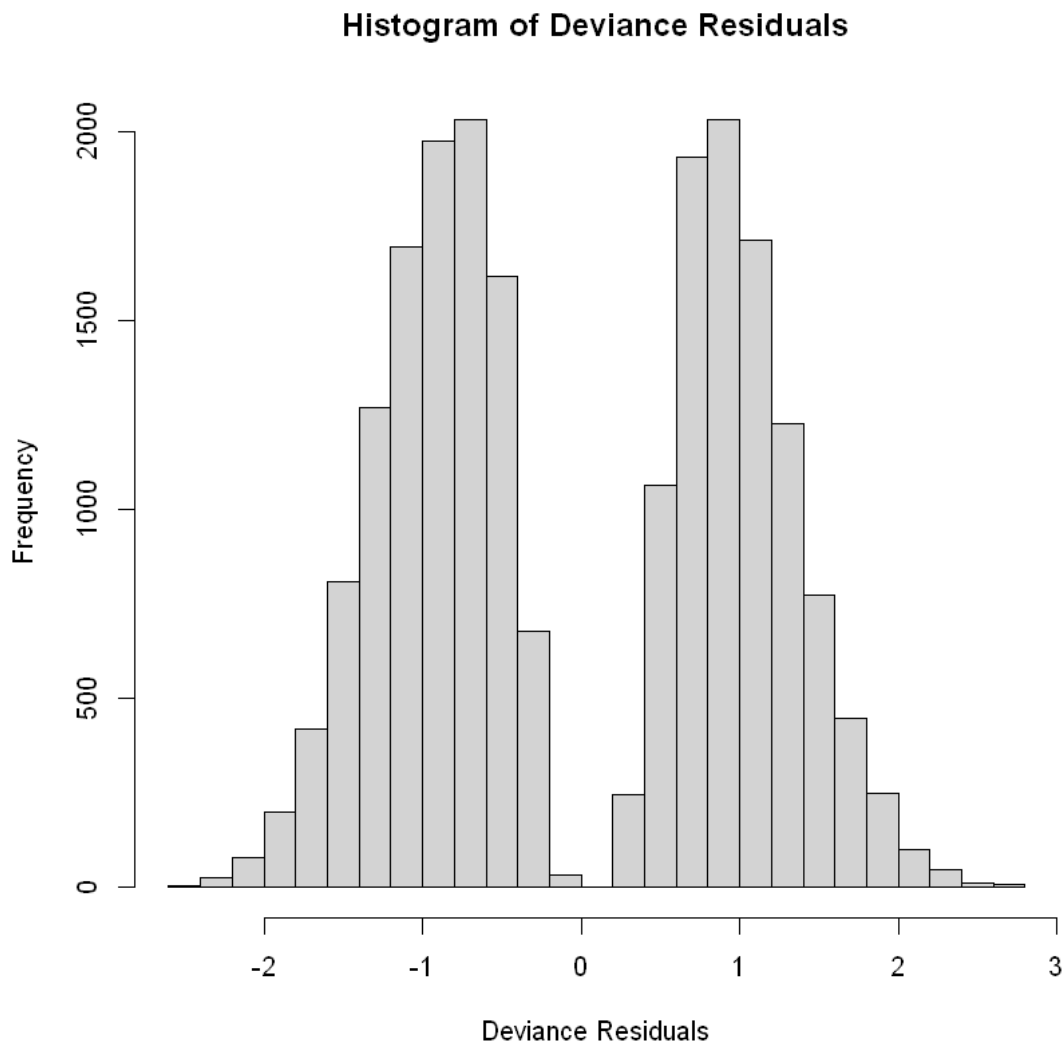
25768

8364 23331

19187

QQ Plot of Deviance Residuals





3 Final Variable Selection

```
[38]: #if we perform variable selection on the transformed variables, could we build  
      ↪ a better model?
```

```
[39]: #build model matrix and response  
X_full_new <- model.matrix(  
  as.formula(paste("Recidivism_Within_2years ~",  
    ↪ paste(selected_vars_reduced_new, collapse = " + "))),  
  data = train_data  
)
```

```

X_new <- X_full_new[, -1] #Remove intercept
Response <- ifelse(train_data$Recidivism_Within_2years == 1, 1, -1)

#define groups (each base variable gets a group number)
col_names <- colnames(X_new)
group <- integer(length(col_names))
group_counter <- 1
group <- as.numeric(factor(group))

#group factor variables (all levels in one group)
for (f in factor_vars) {
  matches <- grep(paste0("^", f), col_names)
  if (length(matches) > 0) {
    group[matches] <- group_counter
    group_counter <- group_counter + 1
  }
}

#group numeric variables (one per group)
for (n in numeric_vars) {
  match_index <- which(col_names == n)
  if (length(match_index) > 0) {
    group[match_index] <- group_counter
    group_counter <- group_counter + 1
  }
}

#standardize the predictors
X_scaled <- scale(X_new)

#cross-validated group lasso
cv_group_lasso_model <- cv.gglasso(
  X_scaled,
  Response,
  group = group,
  loss = "logit",
  pred.loss = "loss",
  nfolds = 10
)

cat("Group Lasso CV Optimized lambda:\n")
print(round(cv_group_lasso_model$lambda.min, 4))

#final group lasso model
group_lasso_model <- gglasso(
  X_scaled,

```

```

    Response,
    nlambda = 100,
    group = group,
    loss = "logit"
)

#extract coefficients at best lambda
coefs <- coef(group_lasso_model, s = cv_group_lasso_model$lambda.min)
selected_groups <- unique(group[which(coefs[-1] != 0)]) # exclude intercept

#map group numbers back to variable names
all_vars <- c(factor_vars, numeric_vars)
selected_vars <- all_vars[selected_groups]
cat("Selected variables (entire groups):\n")
print(selected_vars)

```

Group Lasso CV Optimized lambda:

[1] 2e-04

Selected variables (entire groups):

```

[1] "Race"
[2] "Age_at_Release"
[3] "Violations_ElectronicMonitoring"
[4] "Gang_Affiliated"
[5] "Violations_Instruction"
[6] "Supervision_Level_First"
[7] "Education_Level"
[8] "Dependents"
[9] "Prison_Offense"
[10] "Prison_Years"
[11] "Prior_Arrest_Episodes_Felony"
[12] "Prior_Arrest_Episodes_Misd"
[13] "Prior_Arrest_Episodes_Violent"
[14] "Prior_Arrest_Episodes_Property"
[15] "Prior_Arrest_Episodes_Drug"
[16] "Prior_Arrest_Episodes_PPViolationCharges"
[17] "Prior_Arrest_Episodes_DVCharges"
[18] "Prior_Arrest_Episodes_GunCharges"
[19] "Prior_Conviction_Episodes_Felony"
[20] "Prior_Conviction_Episodes_Misd"
[21] "Prior_Conviction_Episodes_Viol"
[22] "Prior_Conviction_Episodes_Prop"
[23] "Prior_Conviction_Episodes_Drug"
[24] "Prior_Conviction_Episodes_PPViolationCharges"
[25] "Prior_Conviction_Episodes_DomesticViolenceCharges"
[26] "Prior_Conviction_Episodes_GunCharges"
[27] "Prior_Revocations_Parole"
[28] "Prior_Revocations_Probation"
[29] "Condition_MH_SA"

```

```
[30] "Condition_Cog_Ed"
```

```
[31] "Condition_Other"
```

```
[40]: final_formula <- as.formula(paste("Recidivism_Within_2years ~",  
  ↪paste(selected_vars, collapse = " + "))  
final_model <- glm(final_formula, data = train_data, family = "binomial")  
summary(final_model)  
  
#get deviance residuals  
residuals_dev_final <- residuals(final_model, type = "deviance")  
  
#Q-Q plot  
qqPlot(residuals(final_model, type = "deviance"), main = "QQ Plot of Deviance_  
  ↪Residuals")  
  
hist(residuals(final_model, type = "deviance"),  
  main = "Histogram of Deviance Residuals",  
  xlab = "Deviance Residuals",  
  breaks = 20)
```

Call:

```
glm(formula = final_formula, family = "binomial", data = train_data)
```

Coefficients:

	Estimate	Std. Error
(Intercept)	0.1531748	0.1889782
RaceCaucasian	0.0138986	0.0335414
Age_at_Release23-27	-0.4126566	0.0656361
Age_at_Release28-32	-0.8266959	0.0708733
Age_at_Release33-37	-1.1142690	0.0753650
Age_at_Release38-42	-1.3047036	0.0816483
Age_at_Release43-47	-1.4487698	0.0849973
Age_at_Release48 or older	-1.7148975	0.0834821
Violations_ElectronicMonitoring1	0.0065525	0.0644598
Gang_Affiliated1	0.7814890	0.0468649
Gang_AffiliatedUnknown	-0.3346370	0.0496360
Violations_Instruction1	-0.0779053	0.0415331
Supervision_Level_FirstSpecialized	0.0608560	0.0438707
Supervision_Level_FirstStandard	-0.1686272	0.0397616
Supervision_Level_FirstUnknown	0.0343546	0.0669948
Education_LevelHigh School Diploma	0.1088551	0.0449153
Education_LevelLess than HS diploma	-0.0352253	0.0465707
Dependents1	0.0553995	0.0429227
Dependents2	0.0739995	0.0456301
Dependents3 or more	-0.0004496	0.0407550
Prison_OffenseOther	0.1507532	0.0609562
Prison_OffenseProperty	0.1550938	0.0516649

Prison_OffenseViolent/Non-Sex	0.0731450	0.0612043
Prison_OffenseViolent/Sex	-0.3496079	0.1154554
Prison_OffenseUnknown	0.1510692	0.0552179
Prison_YearsGreater than 2 to 3 years	-0.2579005	0.0467021
Prison_YearsLess than 1 year	0.2873224	0.0420048
Prison_YearsMore than 3 years	-0.1830534	0.0481978
Prior_Arrest_Episodes_Felony1	-0.9241454	0.1786870
Prior_Arrest_Episodes_Felony10 or more	-0.0921596	0.1982213
Prior_Arrest_Episodes_Felony2	-0.7083850	0.1786894
Prior_Arrest_Episodes_Felony3	-0.6904069	0.1808301
Prior_Arrest_Episodes_Felony4	-0.4667351	0.1831986
Prior_Arrest_Episodes_Felony5	-0.3991369	0.1860711
Prior_Arrest_Episodes_Felony6	-0.3646381	0.1890241
Prior_Arrest_Episodes_Felony7	-0.5486176	0.1922144
Prior_Arrest_Episodes_Felony8	-0.3200589	0.1963019
Prior_Arrest_Episodes_Felony9	-0.2283998	0.2005381
Prior_Arrest_Episodes_Misd1	-0.0056762	0.0619725
Prior_Arrest_Episodes_Misd2	0.0489381	0.0679669
Prior_Arrest_Episodes_Misd3	0.0313446	0.0747721
Prior_Arrest_Episodes_Misd4	0.1173402	0.0797793
Prior_Arrest_Episodes_Misd5	0.0686495	0.0862784
Prior_Arrest_Episodes_Misd6 or more	0.2594814	0.0824814
Prior_Arrest_Episodes_Violent1	0.0750378	0.0419272
Prior_Arrest_Episodes_Violent2	0.1215185	0.0562806
Prior_Arrest_Episodes_Violent3 or more	0.1209441	0.0646184
Prior_Arrest_Episodes_Property1	0.1189238	0.0515028
Prior_Arrest_Episodes_Property2	0.1397993	0.0621051
Prior_Arrest_Episodes_Property3	0.2206600	0.0738914
Prior_Arrest_Episodes_Property4	0.2402230	0.0852054
Prior_Arrest_Episodes_Property5 or more	0.4390675	0.0891968
Prior_Arrest_Episodes_Drug1	0.0161570	0.0490159
Prior_Arrest_Episodes_Drug2	0.0296322	0.0596423
Prior_Arrest_Episodes_Drug3	0.0263659	0.0696456
Prior_Arrest_Episodes_Drug4	0.0413562	0.0815128
Prior_Arrest_Episodes_Drug5 or more	-0.0386347	0.0826502
Prior_Arrest_Episodes_PPViolationCharges1	0.1529325	0.0536972
Prior_Arrest_Episodes_PPViolationCharges2	0.2727255	0.0613834
Prior_Arrest_Episodes_PPViolationCharges3	0.2627294	0.0684529
Prior_Arrest_Episodes_PPViolationCharges4	0.2960920	0.0754510
Prior_Arrest_Episodes_PPViolationCharges5 or more	0.5098670	0.0733213
Prior_Arrest_Episodes_DVCharges1	0.0692711	0.0506905
Prior_Arrest_Episodes_GunCharges1	-0.0316108	0.0423304
Prior_Conviction_Episodes_Felony1	0.0272476	0.0439248
Prior_Conviction_Episodes_Felony2	0.0027476	0.0544971
Prior_Conviction_Episodes_Felony3 or more	0.0567255	0.0615938
Prior_Conviction_Episodes_Misd1	0.1563572	0.0506524
Prior_Conviction_Episodes_Misd2	0.1974458	0.0602423
Prior_Conviction_Episodes_Misd3	0.1922895	0.0704425

Prior_Conviction_Episodes_Misd4 or more	0.2493507	0.0716998	
Prior_Conviction_Episodes_Viol1	0.0473147	0.0444826	
Prior_Conviction_Episodes_Prop1	0.0139134	0.0475174	
Prior_Conviction_Episodes_Prop2	-0.0214162	0.0639785	
Prior_Conviction_Episodes_Prop3 or more	0.0730162	0.0753222	
Prior_Conviction_Episodes_Drug1	-0.0228546	0.0468998	
Prior_Conviction_Episodes_Drug2 or more	-0.0235598	0.0599680	
Prior_Conviction_Episodes_PPViolationCharges1	-0.1472022	0.0415784	
Prior_Conviction_Episodes_DomesticViolenceCharges1	0.0083270	0.0672161	
Prior_Conviction_Episodes_GunCharges1	0.0095021	0.0547313	
Prior_Revocations_Parole1	0.3415362	0.0536824	
Prior_Revocations_Probation1	-0.1654952	0.0437076	
Condition_MH_SA1	0.2455884	0.0353401	
Condition_Cog_Ed1	0.0319328	0.0341139	
Condition_Other1	0.0806355	0.0382747	
	z value	Pr(> z)	
(Intercept)	0.811	0.417629	
RaceCaucasian	0.414	0.678601	
Age_at_Release23-27	-6.287	3.24e-10	***
Age_at_Release28-32	-11.664	< 2e-16	***
Age_at_Release33-37	-14.785	< 2e-16	***
Age_at_Release38-42	-15.980	< 2e-16	***
Age_at_Release43-47	-17.045	< 2e-16	***
Age_at_Release48 or older	-20.542	< 2e-16	***
Violations_ElectronicMonitoring1	0.102	0.919033	
Gang_Affiliated1	16.675	< 2e-16	***
Gang_AffiliatedUnknown	-6.742	1.56e-11	***
Violations_Instruction1	-1.876	0.060691	.
Supervision_Level_FirstSpecialized	1.387	0.165391	
Supervision_Level_FirstStandard	-4.241	2.23e-05	***
Supervision_Level_FirstUnknown	0.513	0.608095	
Education_LevelHigh School Diploma	2.424	0.015369	*
Education_LevelLess than HS diploma	-0.756	0.449420	
Dependents1	1.291	0.196815	
Dependents2	1.622	0.104862	
Dependents3 or more	-0.011	0.991197	
Prison_OffenseOther	2.473	0.013393	*
Prison_OffenseProperty	3.002	0.002683	**
Prison_OffenseViolent/Non-Sex	1.195	0.232049	
Prison_OffenseViolent/Sex	-3.028	0.002461	**
Prison_OffenseUnknown	2.736	0.006222	**
Prison_YearsGreater than 2 to 3 years	-5.522	3.35e-08	***
Prison_YearsLess than 1 year	6.840	7.91e-12	***
Prison_YearsMore than 3 years	-3.798	0.000146	***
Prior_Arrest_Episodes_Felony1	-5.172	2.32e-07	***
Prior_Arrest_Episodes_Felony10 or more	-0.465	0.641979	
Prior_Arrest_Episodes_Felony2	-3.964	7.36e-05	***
Prior_Arrest_Episodes_Felony3	-3.818	0.000135	***

Prior_Arrest_Episodes_Felony4	-2.548	0.010844	*
Prior_Arrest_Episodes_Felony5	-2.145	0.031947	*
Prior_Arrest_Episodes_Felony6	-1.929	0.053724	.
Prior_Arrest_Episodes_Felony7	-2.854	0.004315	**
Prior_Arrest_Episodes_Felony8	-1.630	0.103008	
Prior_Arrest_Episodes_Felony9	-1.139	0.254730	
Prior_Arrest_Episodes_Misd1	-0.092	0.927022	
Prior_Arrest_Episodes_Misd2	0.720	0.471507	
Prior_Arrest_Episodes_Misd3	0.419	0.675068	
Prior_Arrest_Episodes_Misd4	1.471	0.141343	
Prior_Arrest_Episodes_Misd5	0.796	0.426222	
Prior_Arrest_Episodes_Misd6 or more	3.146	0.001656	**
Prior_Arrest_Episodes_Violent1	1.790	0.073500	.
Prior_Arrest_Episodes_Violent2	2.159	0.030838	*
Prior_Arrest_Episodes_Violent3 or more	1.872	0.061253	.
Prior_Arrest_Episodes_Property1	2.309	0.020939	*
Prior_Arrest_Episodes_Property2	2.251	0.024385	*
Prior_Arrest_Episodes_Property3	2.986	0.002824	**
Prior_Arrest_Episodes_Property4	2.819	0.004812	**
Prior_Arrest_Episodes_Property5 or more	4.922	8.55e-07	***
Prior_Arrest_Episodes_Drug1	0.330	0.741682	
Prior_Arrest_Episodes_Drug2	0.497	0.619307	
Prior_Arrest_Episodes_Drug3	0.379	0.705005	
Prior_Arrest_Episodes_Drug4	0.507	0.611903	
Prior_Arrest_Episodes_Drug5 or more	-0.467	0.640179	
Prior_Arrest_Episodes_PPViolationCharges1	2.848	0.004399	**
Prior_Arrest_Episodes_PPViolationCharges2	4.443	8.87e-06	***
Prior_Arrest_Episodes_PPViolationCharges3	3.838	0.000124	***
Prior_Arrest_Episodes_PPViolationCharges4	3.924	8.70e-05	***
Prior_Arrest_Episodes_PPViolationCharges5 or more	6.954	3.55e-12	***
Prior_Arrest_Episodes_DVCharges1	1.367	0.171766	
Prior_Arrest_Episodes_GunCharges1	-0.747	0.455207	
Prior_Conviction_Episodes_Felony1	0.620	0.535045	
Prior_Conviction_Episodes_Felony2	0.050	0.959789	
Prior_Conviction_Episodes_Felony3 or more	0.921	0.357071	
Prior_Conviction_Episodes_Misd1	3.087	0.002023	**
Prior_Conviction_Episodes_Misd2	3.278	0.001047	**
Prior_Conviction_Episodes_Misd3	2.730	0.006339	**
Prior_Conviction_Episodes_Misd4 or more	3.478	0.000506	***
Prior_Conviction_Episodes_Viol1	1.064	0.287479	
Prior_Conviction_Episodes_Prop1	0.293	0.769670	
Prior_Conviction_Episodes_Prop2	-0.335	0.737821	
Prior_Conviction_Episodes_Prop3 or more	0.969	0.332353	
Prior_Conviction_Episodes_Drug1	-0.487	0.626040	
Prior_Conviction_Episodes_Drug2 or more	-0.393	0.694414	
Prior_Conviction_Episodes_PPViolationCharges1	-3.540	0.000400	***
Prior_Conviction_Episodes_DomesticViolenceCharges1	0.124	0.901408	
Prior_Conviction_Episodes_GunCharges1	0.174	0.862169	

Prior_Revocations_Parole1	6.362	1.99e-10	***
Prior_Revocations_Probation1	-3.786	0.000153	***
Condition_MH_SA1	6.949	3.67e-12	***
Condition_Cog_Ed1	0.936	0.349240	
Condition_Other1	2.107	0.035139	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28605 on 20667 degrees of freedom
 Residual deviance: 25237 on 20583 degrees of freedom
 AIC: 25407

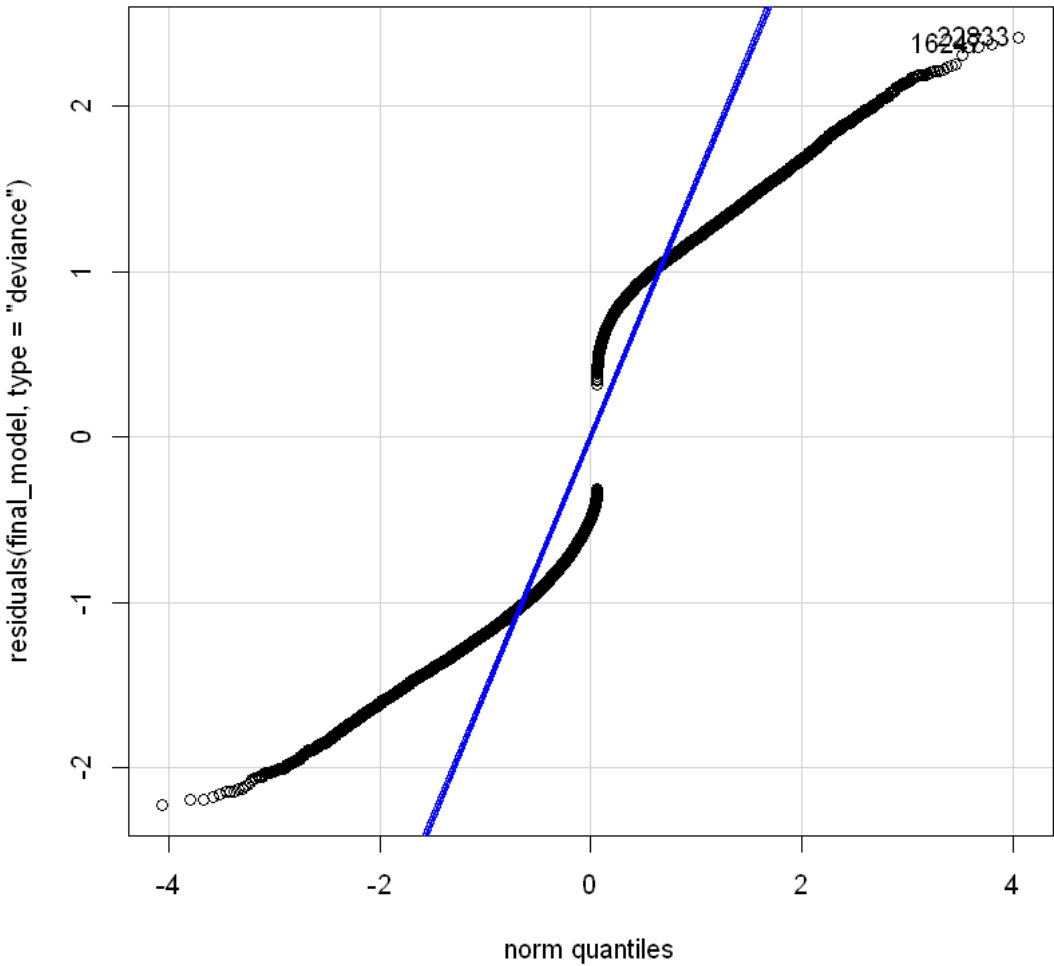
Number of Fisher Scoring iterations: 4

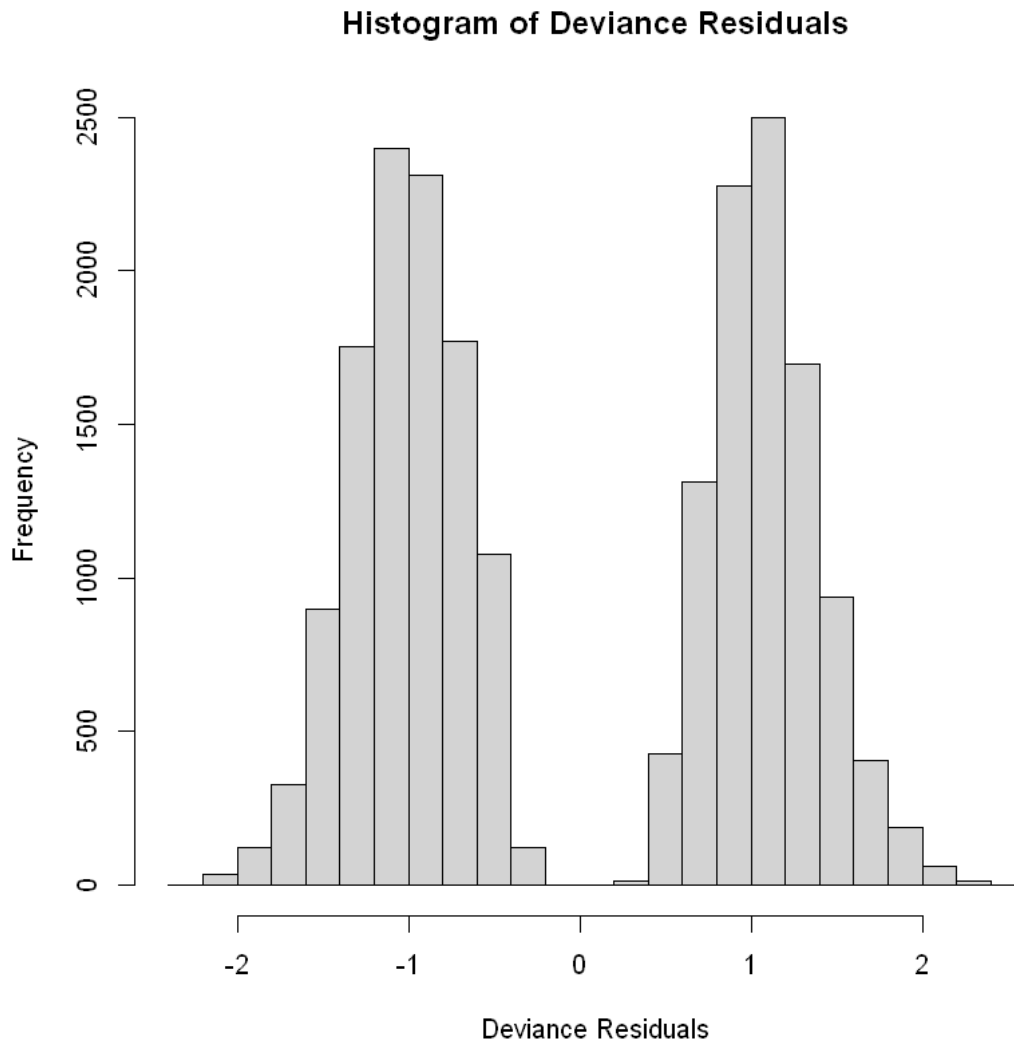
22833

12625 **16247**

20578

QQ Plot of Deviance Residuals





The bimodal residual pattern suggests data might have two distinct subpopulations that respond differently to the predictors. Could be:

- Different risk profiles (e.g., first-time vs. repeat offenders)
- Different demographic groups with different baseline risks
- Missing interaction effects between key variables

```
[41]: #test for overall regression
gstat = final_model$null.deviance - deviance(final_model)
cbind(gstat, 1 - pchisq(gstat, length(coef(final_model)) - 1))
```

A matrix: 1×2 of type dbl $\frac{\text{gstat}}{3368.494 \quad 0}$

```
[42]: with(summary(final_model), c("Residual Deviance" = deviance, "df" = df.
    ↪residual))
pearson_num <- sum(residuals(final_model, type = "pearson")^2)
p_value_deviance <- 1 - pchisq(25237, 20583)
p_value_pearson <- 1 - pchisq(pearson_num, 20583)
p_value_deviance
p_value_pearson
```

```
Residual Deviance          25236.9527250209 df          20583
0
0.33134404095647
```

```
[43]: #improved pearson value, so model may be a slightly better fit.
```

```
[44]: pred_probs_final <- predict(final_model, newdata = test_data, type = "response")
mean(round(pred_probs_final,3))
pred_class_final <- ifelse(pred_probs_final > 0.5, 1, 0)

#confusion matrix
table(Predicted = pred_class_final, Actual = as.numeric(as.
    ↪character(test_data$Recidivism_Within_2years)))

# ROC/AUC
roc_obj_reduced_new <- roc(test_data$Recidivism_Within_2years, pred_probs_final)
auc(roc_obj_reduced_new)
plot(roc_obj_reduced_new)
```

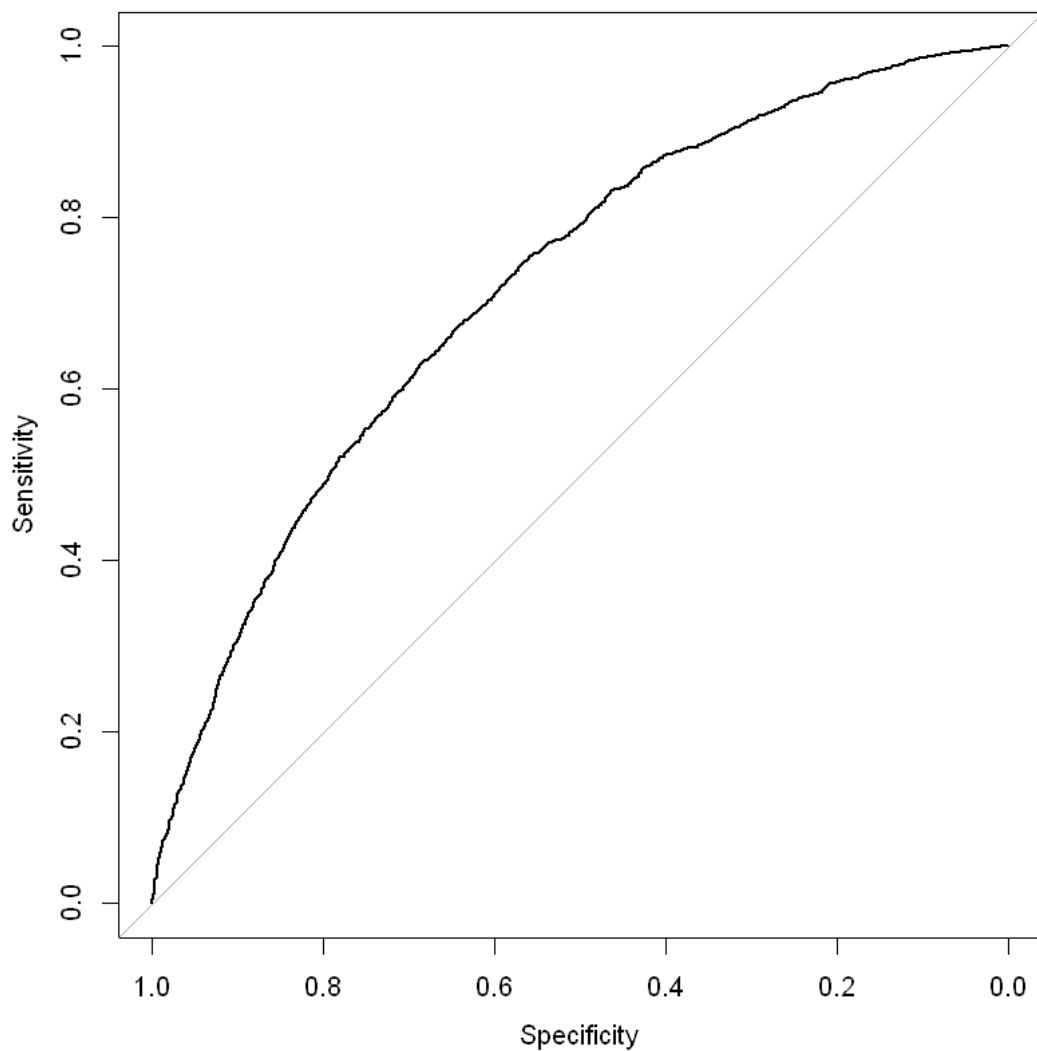
```
0.475086123475905
```

```
      Actual
Predicted  0    1
      0 1872  913
      1  848 1534
```

```
Setting levels: control = 0, case = 1
```

```
Setting direction: controls < cases
```

```
0.718743314142167
```



```
[45]: # Calculate metrics
TN <- 1872
FP <- 848
FN <- 913
TP <- 1534

accuracy <- (TP + TN) / (TP + TN + FP + FN)
sensitivity <- TP / (TP + FN) # Recall, True Positive Rate
specificity <- TN / (TN + FP) # True Negative Rate
precision <- TP / (TP + FP)

cat(sprintf("Accuracy:   %.3f\n", accuracy))
cat(sprintf("Sensitivity: %.3f\n", sensitivity))
```

```
cat(sprintf("Specificity: %.3f\n", specificity))
cat(sprintf("Precision:   %.3f\n", precision))
```

```
Accuracy:    0.659
Sensitivity: 0.627
Specificity: 0.688
Precision:   0.644
```

	reduced_logit_model	reduced_logit_model_new	final_model
Accuracy:	0.711	0.705	0.659
Sensitivity:	0.688	0.692	0.627
Specificity:	0.731	0.717	0.688
Precision:	0.697	0.687	0.644
ROC:	0.784	0.777	0.719

[]:

4 Assessment:

Predictive Performance: * Minimal differences (all within 2%) * Essentially equivalent classification performance (all $\sim .47$) * `reduced_logit_model` better across tmost statistics, but `reduced_logit_model_new` has slightly higher sensitivity

Transformation Worth It: * Better Model Fit: Pearson p-value improved * More Interpretable: Clear employment categories vs. continuous * Stronger Statistical Significance: Employment effects are highly significant * Policy Relevance: “Low/Medium/High employment” is actionable

Final Variable Selection Didn’t Improve Model: * Some important predictive information may have been lost during variable selection — possibly due to group-wise elimination of partially informative features * The selected lambda value may have been too small, which can cause causing under-penalization and allowing too many irrelevant or weak groups to remain

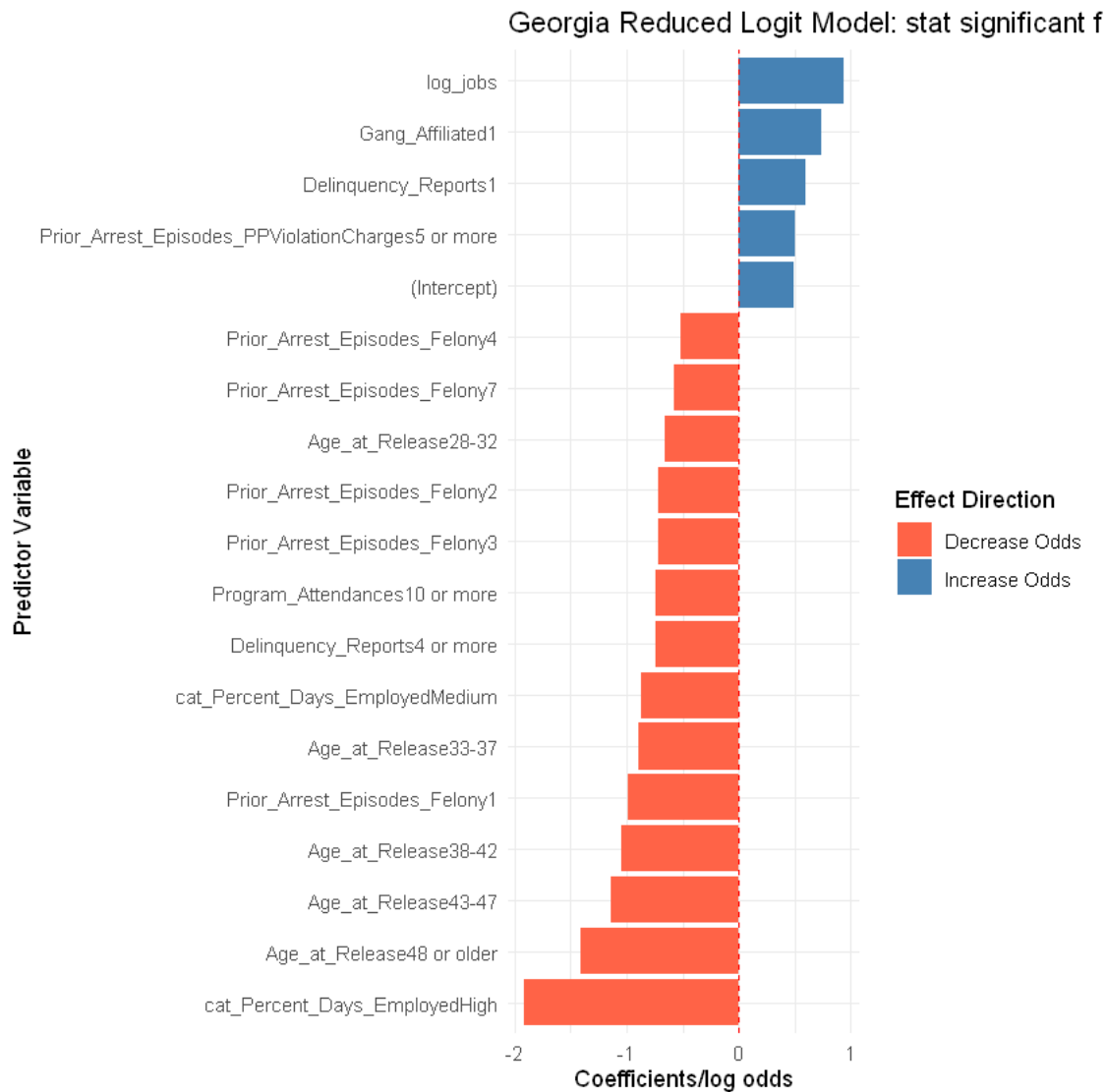
Conclusion: * Variable transformation improved sensitivity * Group lasso did not improve predictive accuracy, potentially because it removed useful features * Best-performing model: Initial logistic regression model * Future iterations could explore post-selection ridge regression, try interactions, or use a different regularization method

5 Ready for exploration! Dataset available as “georgia”. Exploration completed in Georgia - EDA notebook

5.0.1 Note: there are multiple response variables: Recidivism_Within_3years, Recidivism_Arrest_Year1, Recidivism_Arrest_Year2, Recidivism_Arrest_Year3.

6 Visualization of Model Results

```
[48]: #summary(reduced_logit_model_new)
p_values = summary(reduced_logit_model_new)$coefficients
significant = p_values[p_values[,4] <0.05,]
test=as.data.frame(significant) %>% filter (abs(Estimate)>0.45) %>%
  ↪rownames_to_column(var="variable")
#making sure all variables are statistically significant p values so not
  ↪occurring due to chance only
ggplot(test, aes(x = reorder(variable, Estimate), y = Estimate, fill = Estimate,
  ↪> 0)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  coord_flip() +
  scale_fill_manual(
    name = "Effect Direction",
    values = c("TRUE" = "steelblue", "FALSE" = "tomato"),
    labels = c("Decrease Odds", "Increase Odds")
  ) +
  labs(
    title = "Georgia Reduced Logit Model: stat significant for Recidivism",
    x = "Predictor Variable",
    y = "Coefficients/log odds"
  ) +
  theme_minimal()
```



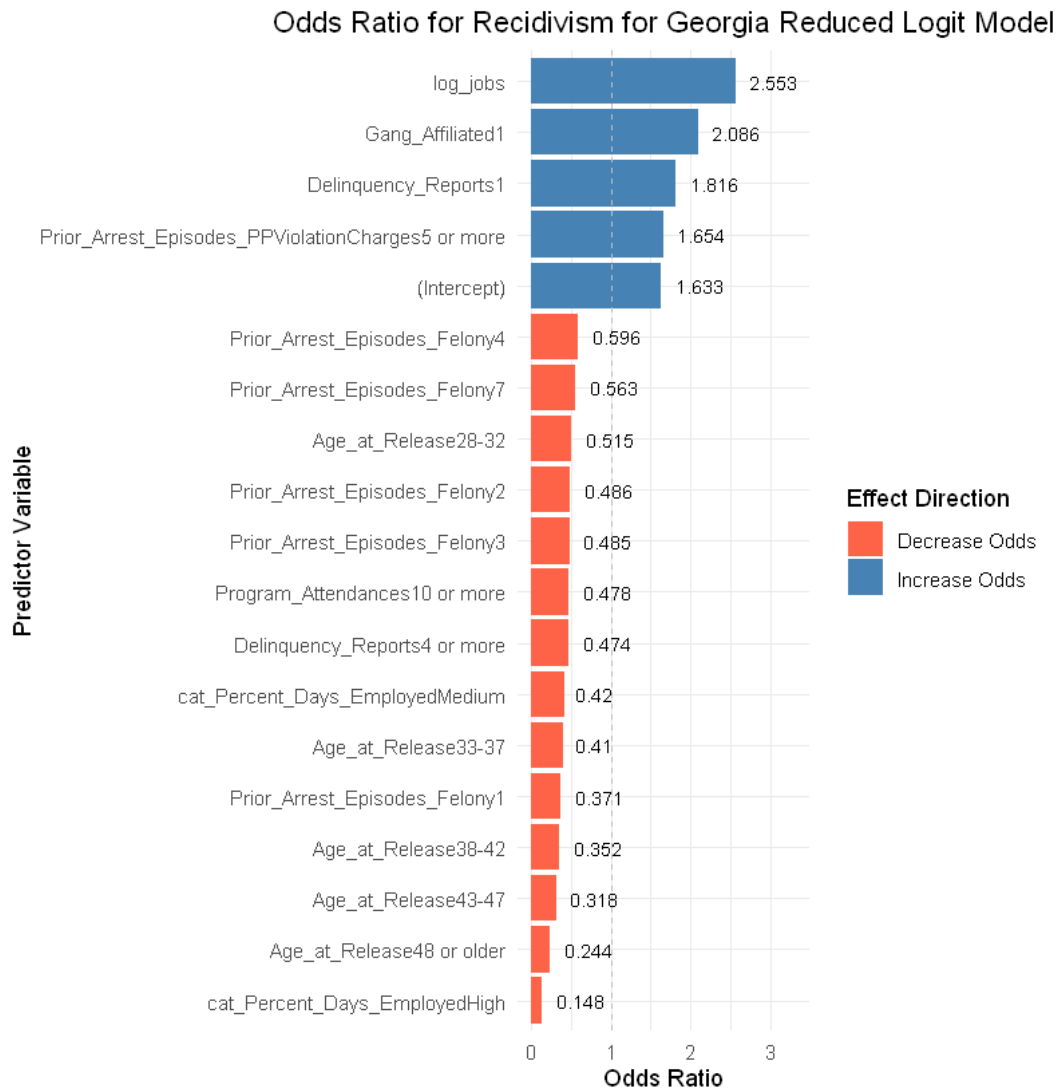
```
[49]: library(tibble)
coef_GA = coef(reduced_logit_model_new)
coef_GA_df = as.data.frame(as.matrix(coef_GA)) %>% rename(log_odds = V1) %>%
  rownames_to_column(var="variable") %>% filter(abs(log_odds)>.45) #filtered
  ↳for variables with larger impacts
#coef_GA_df
```

```
[50]: #as.data.frame(coef_GA)
#coef_GA
```

```
[51]: coef_GA_df = coef_GA_df %>% mutate(odds_ratio = exp(log_odds))%>%
  ↳arrange(desc(odds_ratio)) #calculate odds ratio
```

```
coef_GA_df$probability <- coef_GA_df$odds_ratio / (1 + coef_GA_df$odds_ratio)
  ↪#calculate probability
#coef_GA_df
```

```
[52]: ggplot(coef_GA_df, aes(x = reorder(variable, odds_ratio), y = odds_ratio, fill =
  ↪= log_odds > 0)) +
  geom_col() +
  geom_hline(yintercept = 1, linetype = "dashed", color = "grey", linewidth = 0.
  ↪3) +
  geom_text(aes(label = round(odds_ratio, 3)),
    hjust = -0.3, size = 3, color = "black") +
  coord_flip() +
  expand_limits(y = max(coef_GA_df$odds_ratio) * 1.3) +
  scale_fill_manual(
    name = "Effect Direction",
    values = c("TRUE" = "steelblue", "FALSE" = "tomato"),
    labels = c("Decrease Odds", "Increase Odds")
  ) +
  labs(
    title = "Odds Ratio for Recidivism for Georgia Reduced Logit Model",
    x = "Predictor Variable",
    y = "Odds Ratio"
  ) +
  theme_minimal()+
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.margin = margin(10, 30, 10, 10)
  )
```



```
[53]: ggsave("odds_ratio_georgia.png",
        plot = last_plot(),
        dpi = 300)
```

Saving 6.67 x 6.67 in image

Prior_Arrest_Episodes_PPViolationCharges # Prior GCIC Arrests with Probation/Parole Violation Charges Scale (0-5, where 5=5+) Delinquency_Reports # Parole Delinquency Reports Scale (0-4, where 4=4+)

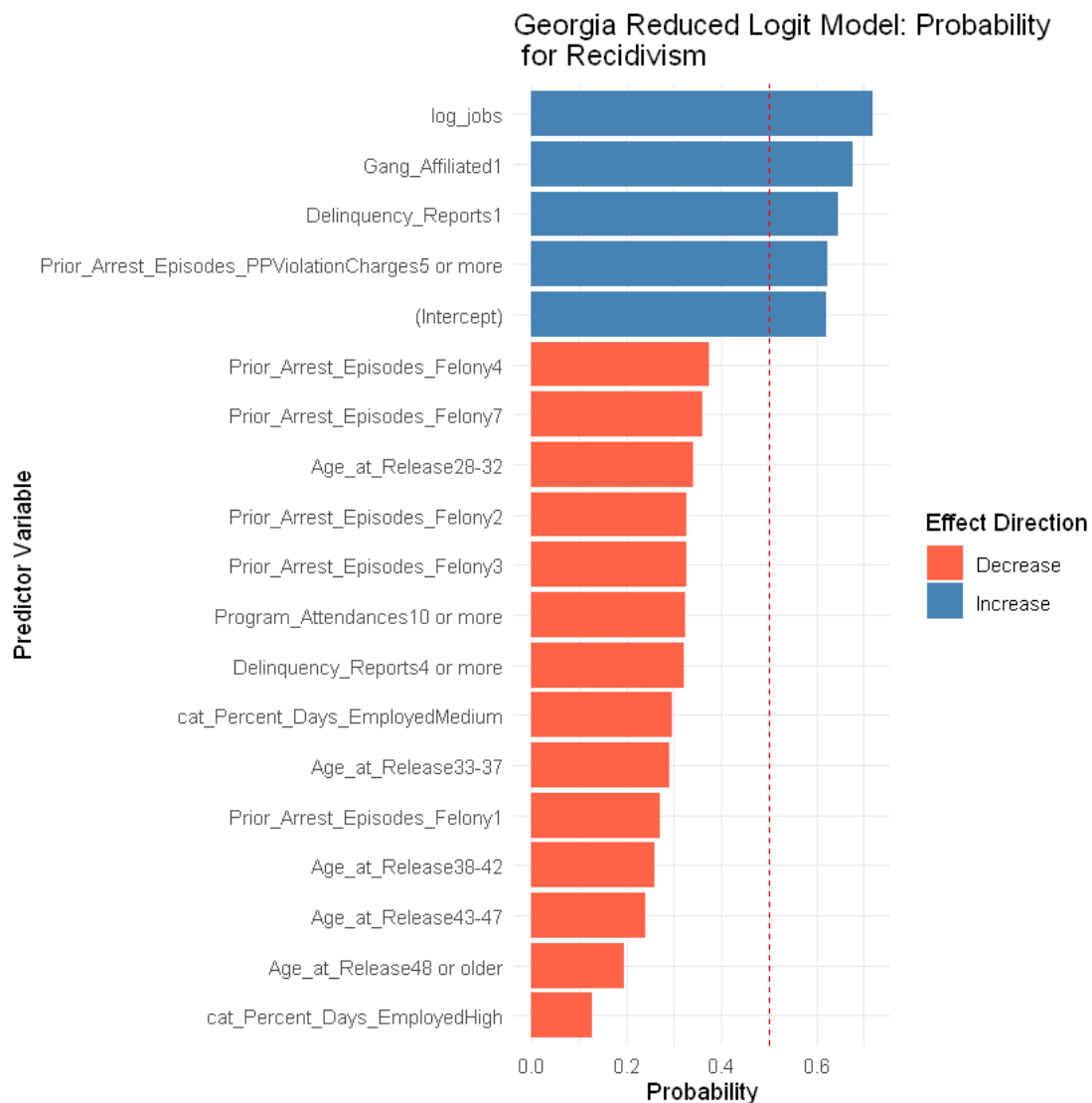
```
[54]: ggplot(coef_GA_df, aes(x = reorder(variable, probability), y = probability,
        ↪ fill = log_odds > 0)) +
        geom_col() +
```

```

geom_hline(yintercept = 0.5, color = "red", linetype = "dashed", linewidth_u
↵= 0.5) +

coord_flip() +
labs(
  title = "Georgia Reduced Logit Model: Probability \n for Recidivism",
  x = "Predictor Variable",
  y = "Probability"
) +
scale_fill_manual(
  name = "Effect Direction",
  values = c("TRUE" = "steelblue", "FALSE" = "tomato"),
  labels = c("Decrease", "Increase")
) +
theme_minimal()

```



```
[55]: #export the cleaned dataframe to a csv file with imputed values populated in
      ↪future notebooks
      #may need to re-initialize factor columns as factors in the next notebook import
      write.csv(georgia, "cleaned Georgia dataset.csv")
```

```
[ ]:
```